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Resilience of dwarf seagrass meadows to simulated marine heat waves

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“This is a story of our changing planet, and what we can do to help it thrive...”

Sir David Attenborough

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Resumo

Desde a Revolução Industrial que o aumento significativo da concentração de dióxido de carbono (CO₂) na atmosfera, resultante das atividades humanas, tem vindo a provocar grandes alterações na físico-química dos oceanos. O CO₂ atmosférico é dissolvido nos oceanos causando uma diminuição do pH da água, através de um processo denominado acidificação dos oceanos. Como resultado do aumento da captação de CO₂ pelo oceano, o pH da superfície oceânica poderá diminuir entre 0,13 e 0,42 unidades, até o final do século XXI. Paralelamente, o aumento do efeito de estufa contribuiu significativamente para o aquecimento global da Terra e subsequentemente dos oceanos, uma vez que estes absorvem mais de 90% da acumulação de calor atmosférico. Deste modo e em resultado da subida das temperaturas médias globais, é previsível que ocorra um aumento de 1°C a 3°C da temperatura da superfície da água do mar (SST), até 2090. Por conseguinte, os eventos climáticos extremos têm vindo a ser cada vez mais frequentes e intensos, tanto a uma escala espacial como temporal. Dentro dos eventos climáticos extremos, as ondas de calor marinhas (MHW) foram identificadas como os principais fatores de *stress* para os organismos marinhos, provocando efeitos prejudiciais sobre as comunidades e ecossistemas costeiros. As MHWs são classificadas de acordo com a sua intensidade, variando de moderadas (categoria I) a extremas (categoria IV) e podem resultar não só de processos da interação entre a atmosfera e o oceano, como também da influência humana que aumenta a severidade e frequência destes eventos. De acordo com o Painel Intergovernamental para as Alterações Climáticas (IPCC), a frequência de MHWs duplicou desde 1982 e tem sido também acompanhada por um aumento de intensidade ao nível mundial. Estudos anteriores revelam que as MHWs provocam efeitos deletérios na biosfera marinha, ameaçando a biodiversidade. Os ecossistemas costeiros são de particular importância uma vez que são zonas cruciais de reprodução para uma grande variedade de organismos marinhos e são dos habitats mais expostos e vulneráveis às alterações climáticas e a consequentes aumentos da temperatura da água do mar. Dentro destes ecossistemas, as pradarias marinhas desempenham um papel ecológico essencial de suporte da biodiversidade uma vez que fornecem alimento, abrigo e habitats berçários para a vida marinha, sendo assim zonas de reprodução muito importantes para peixes e invertebrados. Para além disso, diminuem a erosão costeira e servem de espécies bandeira, indicadoras de estado do sistema onde se incluem. Assim como os mangais e os sapais, as pradarias marinhas são um dos principais sumidouros de carbono azul do planeta, tendo uma elevada capacidade de sequestro de carbono nas suas raízes e rizomas, o que as torna cruciais nas estratégias de combate e mitigação das alterações climáticas. Todas estas características culminam no reconhecimento de serem um dos ecossistemas mais produtivos da biosfera estando esta função ecológica na base todos estes serviços. Deste modo, torna-se imperativo estudar os efeitos que as futuras alterações climáticas poderão provocar nestes ecossistemas.

As ervas marinhas são plantas vasculares com a capacidade de produzir flores, frutos e sementes, pertencendo assim ao grupo das angiospérmicas. Podem ser encontradas nas zonas costeiras de todos os continentes, excetuando a Antártica e crescem em águas pouco profundas devido à necessidade de luz para realização da fotossíntese. Enraízam-se em substratos macios e finos de lama ou areia e em zonas intertidais e subtidais em ambientes marinhos e estuarinos. A diversidade global de espécies de ervas marinhas é baixa (existem cerca de 66 espécies em todo o mundo). No entanto as ervas marinhas podem ter faixas que se estendem por milhares de quilómetros de costa, ocupando entre 160,000 e 600,000 km² a nível mundial. Existem pradarias mono-específicas (formadas por apenas 1 espécie) ou multi-específicas (formadas por várias espécies). As pradarias formadas por um maior número de espécies encontram-se nas zonas dos trópicos, enquanto que nas zonas temperadas as pradarias são formadas por uma ou duas espécies.

No que diz respeito ao impacto das MHWs em ervas marinhas, estudo anteriores mostraram que, quando sujeitas a estes eventos extremos de temperatura há uma diminuição da sobrevivência, do

crescimento e das taxas de fotossíntese destas plantas. As pradarias marinhas têm vindo a desaparecer, não só devido às alterações climáticas, mas também devido a causas relacionadas com a ação humana como técnicas de pesca destrutivas, danos causados pela ancoragem de embarcações de pesca e também por contaminação causada pelo tratamento de resíduos e esgotos urbanos que provocam a eutrofização. Com todos estes impactos, estes habitats costeiros são por isso classificados como vulneráveis e em perigo de extinção na Lista Vermelha de Espécies Ameaçadas da União Internacional para a Conservação da Natureza (IUCN). As pradarias marinhas são conhecidas como *hotspots* de biodiversidade marinha, suportando e aumentando a diversidade da macrofauna associada. Os taxas dominantes da macrofauna associada a estes ecossistemas são os poliquetas, os moluscos e os crustáceos. O estudo das comunidades de macrofauna em pradarias marinhas é um complemento importante a considerar na monitorização destes ecossistemas costeiros por ser um dos elementos de qualidade biológica das águas costeiras de acordo com a Diretiva Quadro da Água da União Europeia.

Zostera noltei encontra-se ao longo da região do Mediterrâneo e do Atlântico Norte Temperado. É uma espécie com um crescimento rápido e uma larga distribuição, formando extensas pradarias em zonas intertidais. Em Portugal, as pradarias de ervas marinhas estão presentes na maioria dos estuários, rias e lagoas. *Z. noltei* é a espécie mais abundante no estuário do Sado, seguindo-se *Cymodocea nodosa* e *Zostera marina* em zonas intertidais e subtidais. A zona de estudo do estuário do Sado é uma área com um impacto acentuado devido ao desenvolvimento agrícola, à presença de infraestruturas urbanas e atividades industriais e portuárias que tem ocorrido nas últimas décadas, tornando-o num sistema de águas poluídas e eutrofizadas com efeitos negativos nas comunidades estuarinas.

A presente dissertação teve como objetivo estudar a resiliência das pradarias da erva marinha *Z. noltei*, e das comunidades de macroinvertebrados associada, no Estuário do Sado a cenários de MHWs de diferentes intensidades. Neste sentido, foram simulados três cenários laboratoriais (*i.e.* controlo, MHW de intensidade III e MHW de intensidade IV) em sistemas de suporte de vida semi-abertos adequados à manutenção *ex situ* de organismos marinhos e no final foram analisadas taxas de sobrevivência, perfil de pigmentos das folhas de *Z. noltei*, diversos parâmetros fotossintéticos e índices de biodiversidade da macrofauna associada (*i.e.* índice de diversidade de Shannon-Wiener, Simpson e Margalef e índice de equitabilidade de Pielou). Não foram observadas diferenças significativas entre tratamentos para qualquer um dos parâmetros analisados das plantas. O mesmo sucedeu com os índices de biodiversidade calculados para a macrofauna associada. A ausência de diferenças significativas entre os vários tratamentos demonstra que as pradarias marinhas de *Z. noltei* no Estuário do Sado estão adaptadas a mudanças abruptas, de curta duração, na temperatura da água. Tal deverá estar associado ao facto desta espécie existir em locais sujeitos a variações ambientais diárias (*i.e.* regimes de marés) nas zonas intertidais. Apesar destes resultados, o estudo do impacto destes eventos climáticos extremos em ervas marinhas continua a ser importante, não só por ser escasso e por haver uma necessidade constante de conservar estes ecossistemas, mas também para que seja possível prever a resiliência destas espécies quando expostas aos cenários futuros propostos pelo IPCC.

Palavras-chave: *Zostera noltei*, Macrofauna, Aquecimento dos Oceanos, Fotobiologia, Pigmentos fotossintéticos

Abstract

Extreme weather events (EWEs) are becoming more frequent and intense, both at a spatial as well as temporal scales. Within EWEs, marine heat waves (MHWs) have been identified as major stressors to marine biota, taunting detrimental effects over coastal communities and ecosystems. According to the Intergovernmental Panel on Climate Change (IPCC), the frequency of MHWs has doubled since 1982 at a worldwide level, being additionally accompanied by an intensity increase. Coastal habitats are of particular concern, and within these habitats, seagrass meadows are known to play an essential ecological role by providing food, shelter and crucial nursery habitats for a wide range of marine species. Additionally, they store staggering amounts of carbon in their roots which makes them important for the carbon cycle. They also act as ecological service providers, by promoting sea bottom stabilization, nutrient cycling, act as buffer/trophic transfer zones to sensitive habitats (*i.e.* coral reefs) and as “environmental status beacons” of coastal ecosystems. As a result of direct anthropogenic pressure and increasing/intensification of MHWs, these key “blue ecosystems” are experiencing a global pressure and subsequent decline. Within this context, the present dissertation aims to investigate the effect of simulated marine heat waves on *Z. noltei* and associated macrofauna, collected in the Sado estuary, by analyzing several endpoints including: survival, photobiological responses, pigment analyses and biodiversity indices (Shannon-Wiener, Simpson and Margalef diversity indexes, and Pielou’s evenness index). No significant effects were observed between treatments for any of the endpoints analyzed. The same happened with the biodiversity indexes calculated for the associated macrofauna. The present findings suggest that *Z. noltei* and associated macrofauna may be resilient to short-term extreme MHW conditions. Yet, it is worth noting that such plasticity and adaptive capacity may not suffice under long-term/chronic warming conditions expected and projected by the IPCC until the end of the century.

Keywords: *Zostera noltei*, Macrofauna, Ocean warming, Photobiology, Photosynthetic pigments

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List of abbreviations and symbols

AF	Absorption factor
AMO	Atlantic multidecadal oscillation
Anthe	Antheraxanthin
Carot/Chl	Carotenoid/chlorophyll ratio
CDR	Climate data record
Chl <i>a</i>	Chlorophyll <i>a</i>
Chl <i>b</i>	Chlorophyll <i>b</i>
CO ₂	Carbon dioxide
DES	De-epoxidation state
DO	Dissolved oxygen
<i>e.g.</i>	‘for example’
ENSO	El Niño–Southern Oscillation
ETC	Electron transport chain
ETR	Electron transport rate
EWE	Extreme weather event
F _v /F _m	Maximum PSII quantum yield
g	gram
GHG	Greenhouse gas
GLM	Generalized linear models
GPS	Gauss-Peak Spectra
<i>i.e.</i>	‘that is’
IPCC	Intergovernmental Panel on Climate Change
IPO	Interdecadal Pacific Oscillation
IUCN	International Union for the Conservation of Nature
LHC	Light-harvesting complex
LSS	Life support systems
MHW III	MHW category III
MHW IV	MHW category IV
MHW	Marine heat wave
NOAA	National Oceanic and Atmospheric Administration
NPQ	Non-photochemical quenching
NSW	Natural seawater
OA	Ocean acidification
PAM	Pulse amplitude modulated
PAR	Photosynthetically active radiation
Pheo <i>a</i>	Pheophytin <i>a</i>
Pheo <i>b</i>	Pheophytin <i>b</i>
PSI	Photosystem I
PSII	Photosystem II
RCP	Representative concentration pathway
SST	Sea surface temperature
T. Auro	Total Auroxanthin
T. Carot	Total Carotenoids
T. Chl	Total Chlorophyll
VDE	Violaxanthin de-epoxidase

Viol	Violaxanthin
ww	wet weight
Zea	Zeaxanthin
β-car	β-carotene
μg	microgram

1. Introduction

1.1. Climate Change

The Earth's climate equilibrium relies on the interaction between physical, chemical and biological processes which occur within the biosphere (Bindoff et al. 2019, IPCC 2019). Since the Industrial Revolution, the concentration of carbon dioxide (CO₂) and other greenhouses gases (GHGs) have been increasing within Earth's atmosphere, mainly as a result of emissions derived from anthropogenic activities, *e.g.* energy production and combustion of fossil fuels (Pachauri & Meye 2014). Although the main drivers of GHG's emissions are human-derived, they also result, at a minor level, from the functioning of the ocean-atmosphere system itself and natural disturbances such as volcanic eruptions (Bindoff et al. 2019). Based upon GHGs concentration trajectories, the United Nations Intergovernmental Panel on Climate Change (IPCC) has adopted the Representative Concentration Pathways (RCPs) in order to describe four possible future climate scenarios, *i.e.* one low emission/stringent mitigation scenario (RCP2.6), two intermediate scenarios (RCP4.5 and RCP6.0) and one high emission scenario (RCP8.5) (Clarke et al. 2014). As a result of upraising CO₂ concentration in Earth's atmosphere, subsequent greenhouse effect, has led to an stepwise temperature increase in the world's oceans, for the last fifty years (Levitus et al. 2000, Epstein & Mills 2005, Gruber 2011). Once water acts as a heat carrier and accumulator (Levitus et al. 2000), the ocean possesses a staggering capacity to accumulate heat (Reichert et al. 2002), absorbing more than 90% of the energy imbalance of the surface of the Earth. The ocean has warmed unabated since 2005, continuing the clear multi-decadal ocean warming trend, particularly in the southern hemisphere (Bindoff et al. 2019). Moreover, this capacity to trap atmospheric heat is foreseen to continue throughout the 21st century (Levitus et al. 2005, Bindoff et al. 2019). Under this context, it is expected that, by the end of 21st century, sea surface temperature (SST) will increase between 1°C (RCP2.6) and 3°C (RCP8.5) (Figure 1.1.1a). Through water circulation and mixing process, heat will be gradually transferred to deeper levels and, depending on the emission scenario, global ocean temperatures are expected to increase between 0.5°C (RCP2.6) and 1.5°C (RCP8.5) (Collins et al. 2013, Abram et al. 2019). Concomitantly, the continuous absorption of atmospheric CO₂ by the world's oceans has given rise to an ongoing process known as ocean acidification (OA), which will ultimately result in a decrease in seawater pH. According to IPCC projections, by the end of the century, oceanic pH will experience an overall decline between 0.13 (RCP2.6) and 0.42 (RCP8.5) pH units in comparison to values dated from the pre-industrial period (Figure 1.1.1b) (Abram et al. 2019).

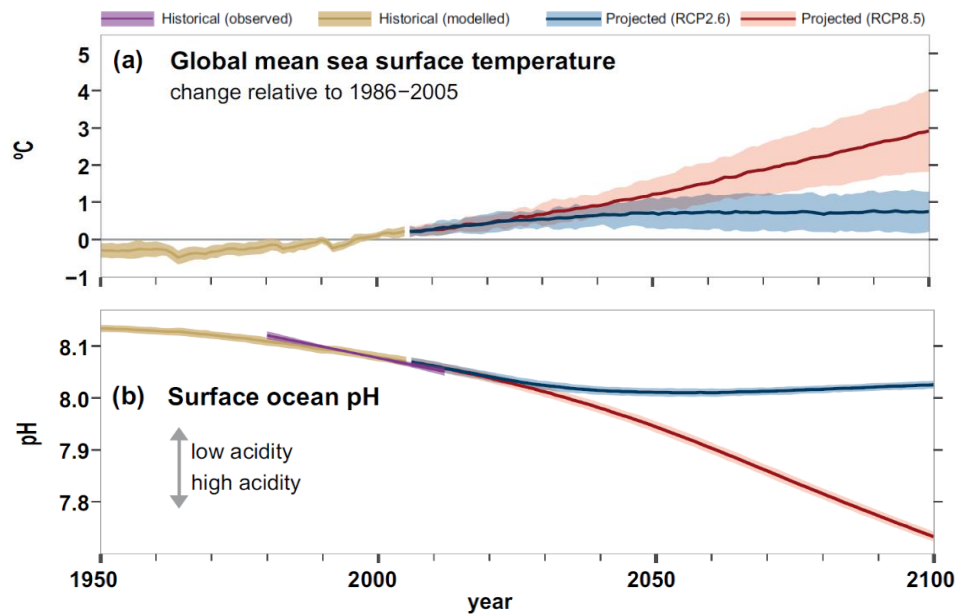


Figure 1.1.1 Projected changes in (a) global mean sea surface temperature and (b) surface ocean pH from observed and modelled records following IPCC low (RCP2.6) and high (RCP8.5) emission scenarios (Abram et al. 2019).

1.2. Extreme weather events and marine heat waves

Extreme weather events (EWEs) are becoming more frequent and intense, both at spatial and temporal scales, pushing earth's climate beyond its natural variability (Rhein et al. 2013, Oliver et al. 2018). An event is considered extreme when it occurs sporadically based on historical data and when it persists for longer than usual, changing the average of the measurement to an extreme itself (Hegerl et al. 2011, Collins et al. 2019). Moreover, these events are also considered extremes based upon the impacts and consequences they pose on ecosystems across the globe (Hegerl et al. 2011). Some examples of meteorological, hydrological and oceanographic-related EWEs include storms, cyclones, floods, droughts, terrestrial and marine heat waves (Herring et al. 2015, Collins et al. 2019) (Figure 1.2.1).

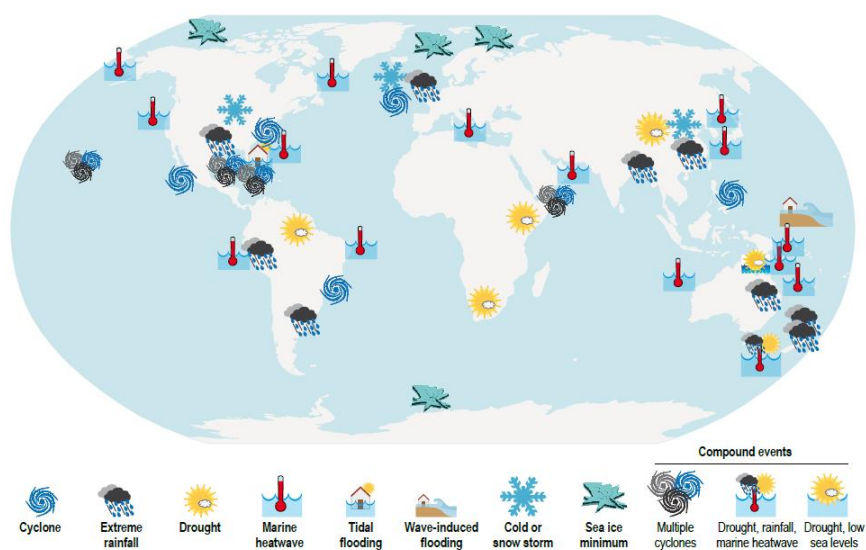


Figure 1.2.1 Overview of global locations where EWE's linked to oceans and cryosphere have been studied (Collins et al. 2019).

Within EWEs, marine heat waves (MHWs) have been identified as a major climate change stressor, posing severe impairments to already high vulnerable coastal and open ocean marine ecosystems (Frölicher et al. 2018). Hobday et al. (2016) describes a MHW as an event taking place in a certain area where there is an extreme increase in SST that exceeds its local 90th percentile value and that lasts for at least 5 consecutive days. The characteristics by which MHWs are classified, such as duration, intensity, rate of onset and their spatial extent, vary widely according to the location of the event (Hobday et al. 2016, Collins et al. 2019). These extreme temperature events (*i.e.* MHWs) have been documented to occur in all world's oceans (Frölicher & Laufkötter 2018), being categorized from moderate (category I) to extreme (category IV), according to their intensity (Hobday et al. 2018) (Figure 1.2.2).

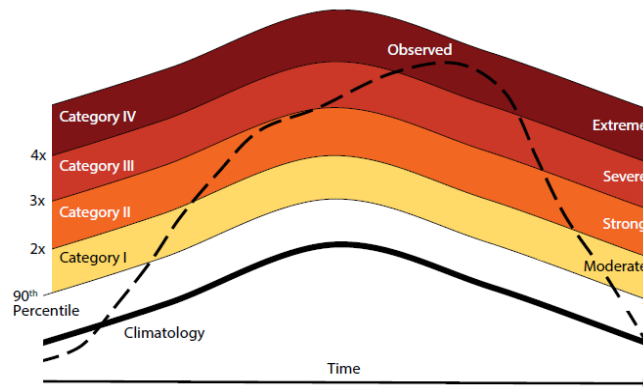


Figure 1.2.2 Categorization of MHWs where the dashed line corresponds to the observed temperature time series, the bold line corresponds to the long-term regional climatology, and the thin line corresponds to the 90th percentile climatological threshold. Multiples of the 90th percentile difference (*e.g.* 2× twice, 3× three times, 4× four times) from the mean climatology value define each of the category (*i.e.* I, II, III and IV) with corresponding descriptors from moderate to extreme (Hobday et al., 2018).

The unusual high temperatures from which MHWs arise derive not only as the result of climate change, but also from oceanographic or atmospheric processes, or their interaction that compromise the marine ecosystems function and structure (Hobday et al. 2016, Frölicher & Laufkötter 2018). Anthropogenic changes also constitute MHW drivers, once they enhance their severity, magnitude, intensity and frequency (Field et al. 2012, Frölicher et al. 2018, Hobday et al. 2018). According to the IPCC, the frequency of MHWs has doubled since 1982 (Collins et al. 2019). Moreover, Frölicher et al. (2018) showed in their study how MHWs are predicted to change over time under different levels of global warming. In both low and high emission scenarios (*i.e.* RCP2.6 and 8.5), global warming will increase the likelihood of MHWs occurring in the near future. According to the low emission scenario (*i.e.* RCP2.6), by the end of the 21st century, there will be an increase in the probability of MHW occurrence by about 20 to 1 but when considering the high emission scenario (*i.e.* RCP8.5) and for the same time period, we are heading for a much higher probability (Figure 1.2.3).

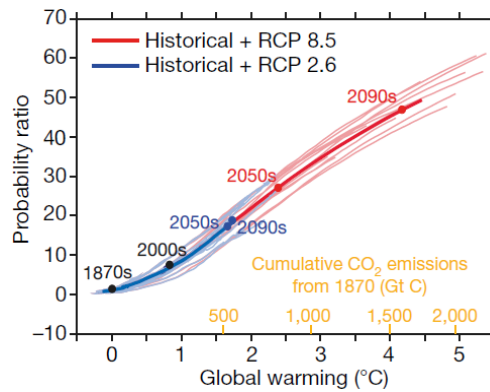


Figure 1.2.3 Global aggregated annual mean probability ratio of MHWs exceeding the 99th preindustrial percentile (Frölicher et al. 2018).

Marine heat waves are normally defined according to SST anomalies. Under this context, there are two characteristic MHW drivers which are relevant in order to understand the causes underlying the occurrence of these events:

- Local processes that influence the temperature budget, where two of them are key elements that modulate MHWs: the Net Surface Heat flux and the current flow (advection) (Figure 1.2.4);
- Climate modes (*e.g.* Interdecadal Pacific Oscillation (IPO), Atlantic multidecadal oscillation (AMO), El Niño–Southern Oscillation (ENSO)) and their teleconnection processes (Holbrook et al. 2019).

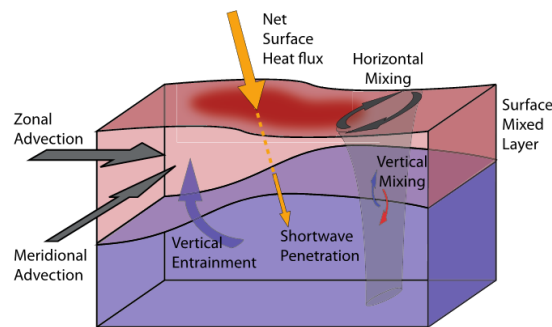


Figure 1.2.4 Local processes/factors that can alter surface mixed layer temperatures (Holbrook et al., 2019 supplementary information).

Although available information on the impacts of MHWs on marine biota and ecosystems is still scarce, several of these events have been reported around the globe during the last years (Holbrook et al. 2019). During 2003, there was a record atmospheric heat wave over Europe, where the Mediterranean Sea SST reached values of 2 to 3°C higher than normal (Spamocchia et al. 2006). The driver of this MHW appeared to be the atmospheric heat wave which occurred at that time (anomalous air-sea heat flux) and it had dramatic effects on rocky reef communities, particularly sponges, soft corals and bivalves (Garrahou et al. 2009). The 2011 MHW in Western Australia reached approximately 3°C above the seasonal value along the western Australian coast. The most probable cause of this EWE was related to a very strong La Nina at the time, besides local changes in the wind pattern which enhanced the warming effect during that particular time (Pearce & Feng 2013). Wernberg et al. (2013) noted significant impacts on biodiversity including the reduction of kelp forests from 80% to 50% cover and the tropicalization of fish communities. The 2012 MHW event in the Northwest Atlantic caused SST to peak 3°C above seasonal values with lobsters moving to coastal and shallower waters making them more easily accessible to harvest (Mills et al. 2013).

The longest MHW on record is “the Blob” which occurred in the Pacific Ocean and lasted for 3 years. This change in water temperature reduced the phytoplankton production changing the oceanic food web and shifting the geographical distribution of some species (Cavole et al. 2016). Before this prolonged extreme event, a MHW in the Tasman Sea region lasted for more than 8 months, whereas a maximum temperature value of +3°C (*i.e.* in comparison to average seasonal values) was reached, as a result of air-sea heat fluxes derived from the atmosphere and from the east Australian ocean current. This event had profound impacts on marine ecosystems and economic activities, with particular emphasis to Pacific oyster, abalone and Atlantic salmon industry (Oliver et al. 2017). Also Smale et al. (2019) investigated some case studies related to MHWs events and their ecological impacts. For instance, these authors described in the Australian region, that the seagrass density loss had a direct relationship as MHW duration (days) per year increased. Also, in California, thermal stress and nutrient limitation associated with a MHW suppressed kelp forest growth, with an observed biomass kelp reduction.

Besides their negative outcomes to marine ecosystems, MHWs can also lead to significant impacts on the human related activities, by directly affecting aquaculture and fisheries practices, besides harvest patterns (Mills et al. 2013). With the lack of information related to the impact of these EWEs on marine biota, there is a need to assess how MHWs will influence ability of marine life to adapt or not to these changing climatic events in the future (Pachauri and Meye, 2014, Frölicher and Laufkötter, 2018).

1.3. Seagrass Meadows: biogeography, species richness, ecosystem services and threats

Seagrasses are angiosperms (flowering plants) that inhabit coastal areas (marine and estuarine environments), where they pioneer some of the most peculiar and important habitats in coastal ecosystems, *i.e.* seagrass meadows (Duarte 2001, Phillips & Milchakova 2003). Estimates of global seagrass coverage area differ greatly throughout the published literature, ranging from 160,000 to 600,000 km² worldwide (Green & Short 2003, Duarte et al. 2010). In a recent study, the authors estimated the potential area suitable for seagrasses to inhabit to be 1,646,789 km² (Jayatilake & Costello 2018). Seagrasses comprise 66 species distributed in 12 genera and 4 families, confined to the marine environment (Larkum et al. 2006). Despite being a group of plants with relatively little diversity of species, they are widely distributed almost all over the world due to their unique features (Orth et al. 2006, Short et al. 2007). They can reproduce under two different strategies: i) vegetative reproduction using clonal rhizome growth and ii) sexual reproduction through flowers and seeds. However, seagrasses devote more of their energy to asexual clonal reproduction. This reproduction strategy is mainly driven whenever there is an environmental disturbance within a seagrass meadow, where clonal rhizome growth will fill out the gaps to sustain the meadow integrity (Phillips & Milchakova 2003, Hogarth 2007).

Seagrasses managed to successfully colonize coastal ecosystems even under the most adverse conditions due to some capacities they developed (Phillips & Meñez 1988, Hogarth 2007). The adaptations include seagrasses ability to withstand varying salinity levels, to grow under complete submersion, to resist to currents and waves influence and stress and finally, to be able to perform submarine pollination (Duarte 2001).

Unlike other coastal habitats such as mangrove forests, salt marshes, coral reefs and kelp beds (Orth et al. 2006), seagrass meadows are geographically distributed along temperate and tropical coastlines in all latitudes (Duarte 2001, Green & Short 2003) with the exception of Antarctica (den Hartog 1970, Green & Short 2003, Waycott et al. 2009), and within six bioregions: Temperate North Atlantic, Tropical Atlantic, Mediterranean, Temperate North Pacific, Tropical Indo-Pacific and Temperate Southern Oceans (Short et al. 2007).

Most species occur in soft muddy or sandy substrates but some species of the genera *Phyllospadix*, *Posidonia* and *Thalassodendron* grow on rocky shelves (den Hartog 1970; Phillips and Meñez 1988; Duarte 2001b; Green and Short 2003). Seagrasses typically grow in marine and estuarine environments with highly variable salinity, ranging from intertidal to subtidal areas (Phillips & Durako 2000, Green & Short 2003), to a maximum of 90 m deep [where *Halophila engelmannii* has been documented (Duarte 1991)]. Aside these reports in deep waters, most seagrass species inhabit shallow waters due to the availability of light for photosynthesis (photosynthetic active radiation), which diminishes with increasing water depth (Phillips & Durako 2000, Green & Short 2003, Hogarth 2007). Most of seagrass beds are monospecific, but there are also multispecific communities, where up to 12 species could be present in the same meadow (Duarte 2001, Hogarth 2007).

Seagrass meadows exhibit a biogeographic pattern of latitudinal biodiversity, suggesting that seagrass species richness decreases from the equatorial zone to more temperate and colder zones. In the tropics there is a tendency for mixed meadows to appear, whereas in the poles there are more seagrass beds which are monospecific (Duarte 2001, Phillips & Milchakova 2003) (Figure 1.3.1).

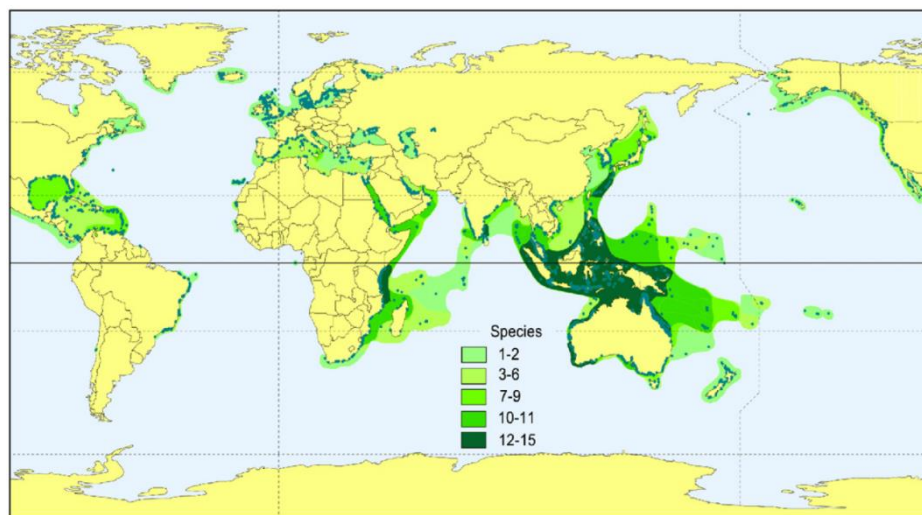


Figure 1.3.1 Global distribution and diversity of seagrasses worldwide. Shades of green correspond to the number of species reported for an area (Short et al. 2007).

Seagrass meadows represent one of the most productive and important ecosystems, providing food, shelter and a nursery ground for a wide variety of invertebrate and vertebrate species (Beck et al. 2001, Green & Short 2003), ranging from fungi to mammals (Duarte 2001). Additionally, seagrass meadows act as ecological service providers, by promoting sea bottom stabilization, reducing coastal erosion, promoting nutrient cycling and acting as buffer/trophic transfer zones between habitats (e.g. coral reefs, saltmarshes and mangroves) (Beck et al. 2001, Green & Short 2003, Phillips & Milchakova 2003). Within these habitats, seagrasses act as “active environmental status beacons” due to their sensitivity to environmental deterioration (Orth et al., 2006).

Furthermore, seagrass meadows also take up the role as biological indicators of water (Phillips & Durako 2000, Phillips & Milchakova 2003) and overall ecosystem quality, following the European Water Framework Directive [WFD, 2000/60/EC (European Commission 2000)] and the Marine Strategy Framework Directive [MSFD, 2008/56/EC (European Commission 2008)].

Seagrass meadows play a key role within the so called ‘blue carbon’ habitats (Duarte et al. 2005, McLeod et al. 2011), since overall they represent one of the main sources of photosynthetic primary production, playing an important role in the carbon cycle, namely through their capture and storage (Hogarth 2007, Bindoff et al. 2019). Within these habitats, seagrasses possess high rates of carbon

removal ($138 \pm 38 \text{ g C m}^{-2} \text{ yr}^{-1}$) (McLeod et al. 2011), storing nearly 20% of blue carbon in their roots and rhizomes (Short et al. 2018). However, when these ‘blue carbon’ ecosystems are disturbed, they release part of their stored carbon back to the atmosphere (Marbà & Duarte 2010, Arias-Ortiz et al. 2018). Therefore and given the importance that seagrasses have to the global oceanic carbon budget (Duarte et al. 2005), seagrass beds form important habitats to mitigate climate change (Arias-Ortiz et al. 2018) and so it is necessary to enforce their protection. Considering all these unique characteristics, seagrasses meadows are believed to be some of the most valuable marine ecosystems worldwide, providing numerous important ecosystem goods and services (Costanza et al. 1997).

However, seagrass meadows are rapidly declining at a worldwide level, mainly as a result of human activities, such as the ones related to dredging, vessel anchoring, aquaculture (*e.g.* land-based fish farms) and bivalve harvesting (Green & Short 2003, Hogarth 2007). It has been estimated that since 1980 the coverage area of these habitats has been decreasing at a rate of 110 Km^2 per year (Waycott et al. 2009). According to the International Union for the Conservation of Nature (IUCN) Red List of Threatened Species (IUCN 2021), nearly 20% of seagrass species are at risk of extinction, being listed as endangered, vulnerable or near threatened (Short et al. 2011). Indirectly, other human activities such as agriculture practice, industry and waste landfills could cause severe impairments to these ecosystems health, through water bodies eutrophication (*i.e.* excess nutrient input), or even organic and inorganic pollution. Although these human-induced disturbances are considered to be the main causes for the widespread depletion of these habitats (Hogarth 2007), they are also highly threatened by climate change-related drivers (Short & Neckles 1999, Duarte et al. 2008, Waycott et al. 2009) such as MHWs, which directly affecting survival, growth, photosynthesis and respiration, besides seagrass flowering and seed germination (Short et al. 2001). These extreme events have already led to an increase in seagrass shoot mortality and spatial coverage contraction in the Mediterranean Sea (Díaz-Almela et al. 2009, Marbà & Duarte 2010) in the southwestern Baltic Sea (Reusch et al. 2005) and also in Western Australia (Smale et al. 2019).

1.4. Seagrass-associated macrofaunal assemblages

It is well established that seagrass meadows increase macrofaunal biodiversity and the abundance of species, both benthic infaunal and epifaunal species (Boström et al. 2006, Leopardas et al. 2014). Epifaunal species inhabit among plant leaves and stems and, therefore are more dependent on the structure of seagrasses (*e.g.* shoot density and leaf morphology). Infaunal species live in the bottom sediments (Orth et al. 1984, Tanner 2005). Among seagrass-associated macrofaunal communities, polychaetes, molluscs and crustaceans are the dominant taxa (Adulyanukosol & Poovachiranon 2006). Previous studies have shown that macrofaunal density and species richness is considerably higher in vegetated seagrass areas when compared with unvegetated areas. This difference is due to the fact that seagrasses provide food, habitat and shelter, besides providing protection from predation (Stoner 1980, Lewis & Stoner 1983).

In seagrass meadows, macrofaunal functional diversity (*i.e.* number of functional groups) is a critical issue. In species-poor areas there will be few species per functional group, resulting in an ecosystem more susceptible to disturbances than ecosystems with multiple species playing the same functional role (Duffy et al. 2001, Barnes & Hendy 2015). Furthermore, seagrass macrofaunal assemblages are generally composed by generalist species that can thrive in various environments, which also makes them more resilient (Boström et al. 2006). Benthic macrofauna play an important ecological role in a detritus-based food web of seagrass, because they are the primary link in the trophic chain between primary producers and higher trophic-level predators (Orth et al. 1984).

Studying macrofaunal species are of paramount importance, specially within the framework of the European Water Framework Directive [WFD, 2000/60/EC (European Commission 2000)], that says

that marine macrofauna is one of the most important biological indexes used to monitor coastal waters. Besides that, when studying the impact of any stressor in seagrasses, it is important to consider that a decrease in seagrass meadows will elicit cascading effects at population, community and ecosystem levels. Therefore, the assessment of seagrass-associated macrofauna is an important conservation and management strategy to monitor seagrass meadows (Reed & Hovel 2006).

1.5. *Zostera noltei* and the Sado estuary

The seagrass *Zostera noltei* (Homemann) is found along the Mediterranean and Temperate North Atlantic bioregions (Short et al. 2018). It is one of the most widespread seagrass species that inhabits the European and western Africa coastlines (Green & Short 2003, Marbà et al. 2013), besides the Caspian and Aral Seas in western Asia (Phillips & Meñez 1988, Larkum et al. 2006). This fast-growing foundation species (Larkum et al. 2006) inhabits soft substrates, within estuarine and sheltered environments. It forms extensive meadows mainly in the intertidal zone, which make this species particularly susceptible and vulnerable to climate change related events (Short & Neckles 1999) and anthropogenic pressures (Duarte et al. 2008).

Despite *Z. noltei* being able to tolerate temperatures of up to 37°C for an extended time period, i.e. 21 days (Massa et al. 2009), increasing SST is expected to shift *Z. noltei* range and distribution towards the polar regions, by the end of this century (Valle et al. 2014). Although this species is considered as least concerned (IUCN 2021), *Z. noltei* has a declining population trend (Short et al. 2011). Yet, in Europe, there has been an increase in its coverage during the 2000s (de los Santos et al. 2019). Along the Portuguese coast it is possible to find three seagrass species: *Zostera noltei*, *Zostera marina* and *Cymodocea nodosa* (Cunha et al. 2009). *Zostera noltei* is the most widespread species (15.74 Km²) occurring in Sado, Mira, Guadiana, Mondego, Tagus and Arade estuaries, besides Ria Formosa, Ria de Aveiro and other coastal lagoons (Cunha et al. 2013). According to Cunha et al. (2013), Ria Formosa has the highest seagrass coverage (15 Km²) followed by Ria de Aveiro (0.56 Km²) and the Sado estuary (0.33 Km²).

The Sado estuary is the second largest in Portugal with an area of approximately 24,000 ha, being located in the West Coast of Portugal (i.e. Setúbal Peninsula). In addition to the natural resources, the majority of the Sado Estuary is classified as an important Nature Reserve, the Sado Estuary Natural Reserve (D. L. n.º 430/80) and it is internationally protected by the Ramsar Convention due to high biodiversity values, natural functions, fauna and flora richness and diversity of habitats (Caeiro et al. 2002, Coutinho 2003).

Within Sado Estuary, it has been identified 3 seagrass species (*Zostera marina*, *Z. noltei* and *Cymodocea nodosa*) that cover 0.33 Km². The dominant species is *Z. noltei* with approximately 0.28 Km² (203 meadows), followed by *C. nodosa* (0.03 Km²) and *Z. marina* (0.01 Km²) covering intertidal and subtidal areas (Cunha et al. 2009, 2013).

Besides that, this transition environment possesses a strong presence of urban infrastructure, industrial and harbor activities which directly or indirectly pose negative effects on associated biological communities within the estuary. The Sado estuary has also been identified as an environment with a large input of organic enrichment from industrial and urban disposal during the last decades, making it a system also affected by eutrophication (Rodrigues 1992, Coutinho 2003). This is critical information to consider when studying coastal ecosystems because a recent study has shown that pollution caused by poor wastewater treatment coupled with MHWs were the cause of losses in the richness of benthic communities in south Brazil (Brauko et al. 2020).

2. Objectives

The overall aim of the present dissertation was to investigate the effect of marine heat waves on *Z. noltei* and associated macrofauna, collected in the Sado estuary. More specifically, I aimed to explore the effects of different categories of simulated (under laboratorial conditions) marine heat waves on the seagrass *Z. noltei* and the associated macroinvertebrate communities, by analyzing several experimental endpoints namely survival, photobiological responses, pigment analyses and biodiversity indices (Shannon-Wiener, Simpson and Margalef diversity indexes, and Pielou's evenness index). Understanding how *Z. noltei* and associated macroinvertebrate fauna will respond to extreme weather events is of paramount importance in order to better understand species resilience to upcoming climate change stressors, *i.e.* MHWs.

3. Materials and Methods

3.1. Seagrass collection

Zostera noltei patches were collected (June 2020) in the Sado estuary, at Eurominas area (38°28'31" N; 8°46'34" W, Setúbal, Portugal), alongside with its root material and associated sediment (Figure 3.1.1). Upon collection, each patch, from now on designated as experimental replicate (n=23 in total; n=18 for experimental procedures and n=5 for initial shoot density assessment) was allocated into an individual tank (23.5 x 14.5 cm; Length x Width) and covered with seawater to avoid air desiccation. Subsequently, all tanks were transported to the aquatic facilities of the Laboratório Marítimo da Guia (Cascais, Portugal).

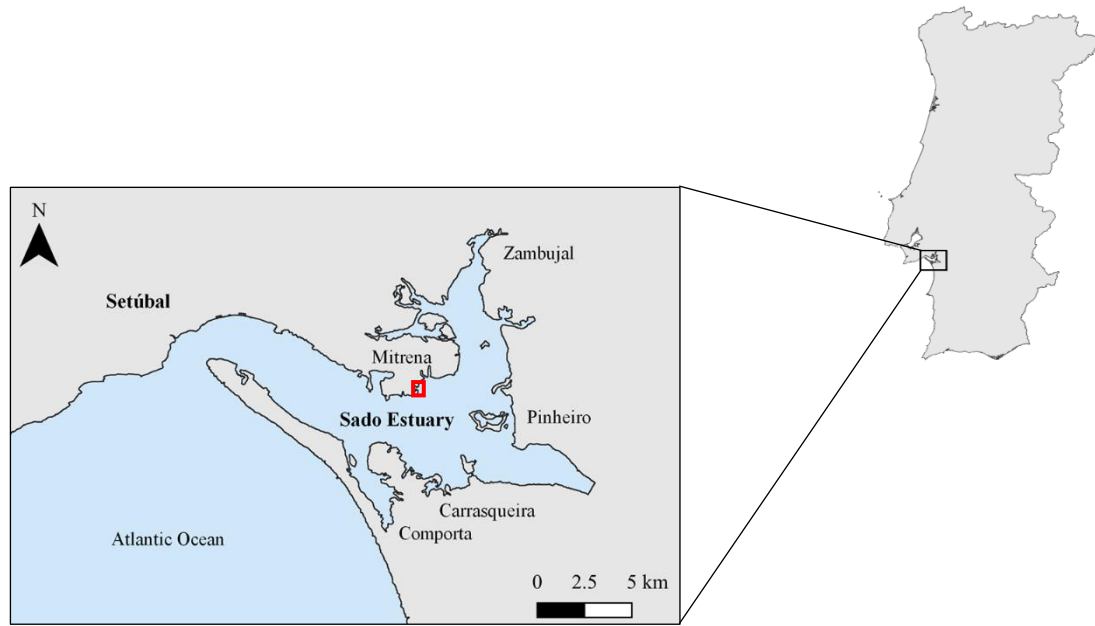


Figure 3.1.1 Geographic location of the Sado estuary and the sampling location (i.e. red square). Map obtained using the geographic information software QGIS 3.4.8 (North River Geographic Systems, Inc., USA).

3.2. Laboratory acclimation

Experimental replicates (n=18) were randomly assigned, and laboratory acclimated for 12 days within semi-open aquatic life support systems (LSS, n=3, Figures 3.2.1 and 3.2.2). Natural seawater (NSW) was pumped directly from the sea and subsequently mechanically filtered (down to 5 μm) and UV sterilized (Vecton 120, TMC Iberia, Portugal), before being supplied to each LSS. During laboratory acclimation, a daily monitoring of seawater temperature ($18.2 \pm 0.3^\circ\text{C}$, mean $\pm$ SD), pH (8.143 ± 0.037 , mean $\pm$ SD), dissolved oxygen (DO) ($9.42 \pm 0.63 \text{ mg L}^{-1}$, mean $\pm$ SD) and salinity ($35.2 \pm 0.3 \text{ ppt}$, mean $\pm$ SD) was performed. Each experimental replicate was continuously supplied with pre-conditioned NSW, at a flow rate of $3.96 \pm 1.21 \text{ mL s}^{-1}$ (mean $\pm$ SD). Overhead artificial illumination (Chihiros LED A601, Zhejiang, China) was provided and photoperiod was set to 14h:10h (light/dark cycle). Photosynthetically active radiation (PAR) was measured at the beginning of the acclimation period, using a portable fluorometer and maintained at 310 ± 27 (mean $\pm$ SD) $\text{mol photons m}^{-2} \text{ s}^{-1}$ (Silva et al. 2013).

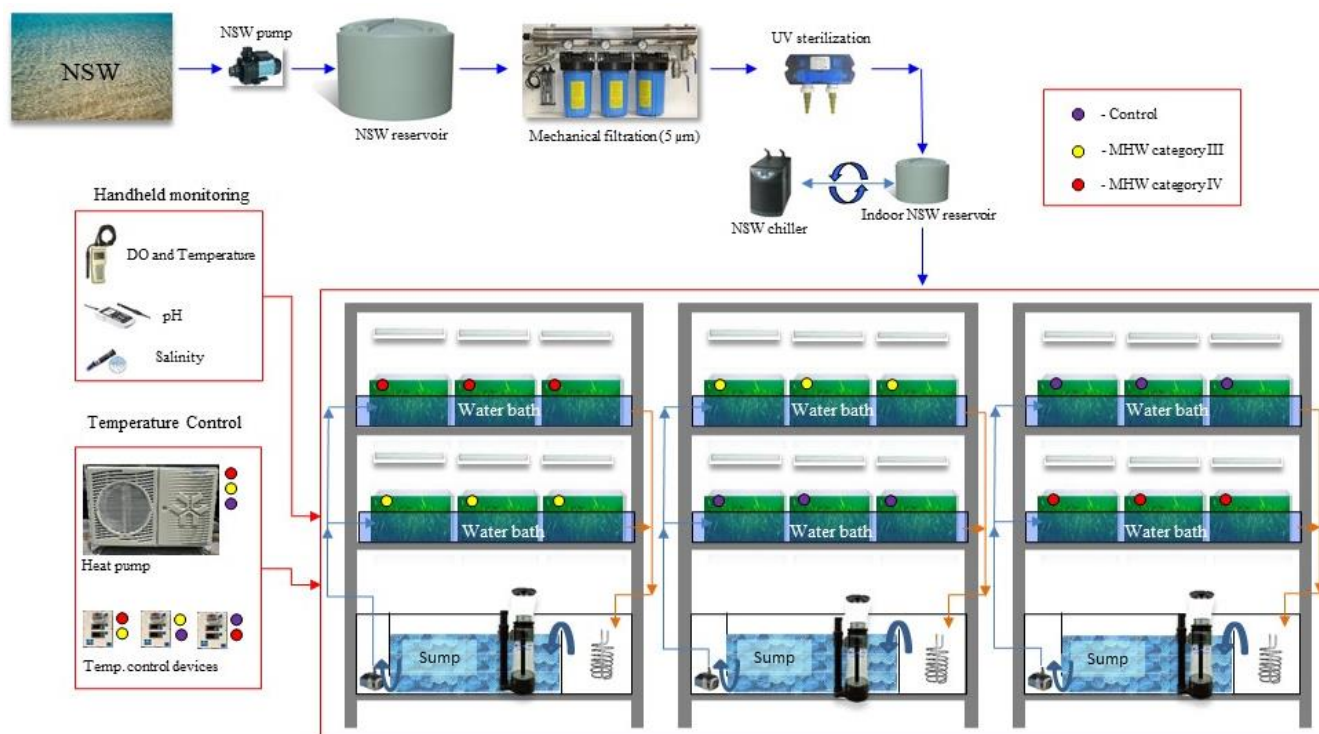


Figure 3.2.2 Schematic representation of LSS where exposure of seagrasses to acclimation and experimental conditions (*i.e.* control, MHW categories III and IV) was performed.

3.3. Experimental Exposure

Following the acclimation period, experimental replicates were exposed to 3 experimental conditions (n=6 to each condition), comprising a control and 2 MHW scenarios (*i.e.* categories III and IV events). A 36 years (1982 - 2018) dataset for SST in the region of Sado estuary (Setúbal, Portugal) was acquired from the program Climate Data Record (CDR) of the National Oceanic and Atmospheric Administration (NOAA) (Banzon et al. 2016). The R package, *heatwaveR* (Schlegel & Smit 2018) was used to determine the average duration (14 days) and the maximum temperatures registered (21.6 and 22.7°C, corresponding to a MHW categories III and IV respectively), when the climatology was ~18.4°C. This package calculates and displays MHWs according to the definition by Hobday et al. (2018). As a result, projections were made for the different experimental scenarios (Figure 3.3.1). Within each LSS, experimental replicates were 2/3 submerged in the water column. Custom built temperature-controllers were connected to submerged thermostats (200W, TMC Iberia, Portugal) and water circulating pumps (TMC Iberia, Portugal) in order to control temperature (to the nearest $\pm 0.2^\circ\text{C}$). Each LSS was equipped with a protein skimmer (V2skim Pro 450, TMC Iberia, Portugal), mechanical (200 μm , TMC Iberia, Portugal) and biological filtration (Ouriço® bioballs, Fernando Ribeiro, Portugal).

Temperature and DO were monitored using a portable device (Multi 3510 IDS, WTW, Germany) connected to an optical IDS DO sensor (FDO®925, WTW, Germany). Monitoring of pH values was performed using a portable device (pHenomenal® pH 1100 H, VWR, USA) connected to an epoxy pH electrode (DJ 113, VWR, USA). Salinity was measured using a handheld refractometer (V2 refractometer, TMC Iberia, Portugal). Seawater parameters (temperature, pH, DO and salinity) were daily monitored and recorded (Supplementary Table 1).

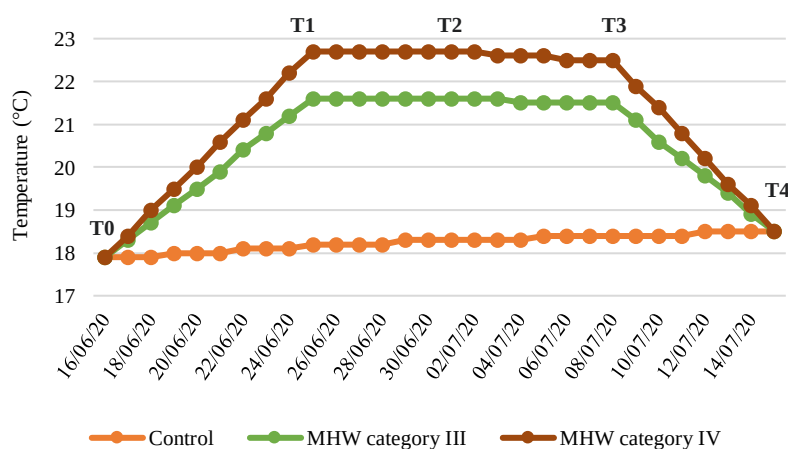


Figure 3.3.1 Daily temperature setup of experimental conditions (control, MHW categories III and IV). T0-T1: initial temperature ramp (MHW categories III and IV); T1-T3: temperature plateau (MHW categories III and IV); T3-T4: recovery period.

3.4. Shoot density and Gauss - Peak Spectra pigment analysis

Zostera noltei shoot density was recorded after field sampling (T0) and at the end of the exposure to experimental conditions (T4). This endpoint was expressed as the number of shoots per area (shoots cm^{-2}). For the Gauss - Peak Spectra pigment analysis, *Z. noltei* leaves were cut and weighed at the end of the recovery period (T4). Samples were immediately frozen (liquid nitrogen) and stored at -80°C , until pigment analyses were performed. To perform pigment extraction, leaves were freeze-dried in the dark for 48 hours with acetone (99.99%, Fisher Chemical) followed by vegetable material fragmentation using a handheld grinder (T10 basic ultra-turrax, IKA, Germany) and an ultrasonic bath

(Duarte et al. 2015). Acetone was the solvent used for pigment extraction as it prevents the artifact formation of metal substituted chlorophylls ([Hms]-Chls) (*i.e.* exchange of the central Mg^{2+} ion by other metal ions, mainly Zn^{2+} and Cu^{2+}) (Küpper et al. 2006). Extraction was dark performed (avoidance/prevention of pigment degradation) in 24 hours at $-20^{\circ}C$. Following pigment extraction, samples were centrifuged (3040g, 15 minutes, $4^{\circ}C$) to separate samples into a gradient of density and allow the collection of the supernatant for posterior analysis. Since the samples were diluted in acetone and prior to scanning them in a dual beam spectrophotometer (UV-1800, Shimadzu, Japan), a baseline was established which allowed acetone absorbance to be removed from each reading. An absorbance spectrum (350 nm to 750 nm) in 0.5 nm steps was selected. The resulting absorbance spectra was processed using the SigmaPlot Software (Systat Software, Inc., USA) and using the Gauss-Peak Spectra (GPS) fitting library method (Küpper et al. 2007), allowing the identification and quantification of pigments (chlorophyll *a*, chlorophyll *b*, pheophytin *a*, pheophytin *b*, auroxanthin, antheraxanthin, violaxanthin, zeaxanthin, β -carotene and lutein). In order to better evaluate the protection efficiency of the photosynthetic apparatus, De-Epoxidation State (DES) of xanthophyll cycle pigment was calculated as:

$$DES = \frac{[Antheraxanthin] + [Zeaxanthin]}{[Violaxanthin] + [Antheraxanthin] + [Zeaxanthin]}$$

The xanthophyll cycle is a cyclic reaction that comprises two successive de-epoxidations (removal of two oxygen functions) of violaxanthin via antheraxanthin and zeaxanthin, catalyzed by violaxanthin de-epoxidase (VDE) (Demming-Adams & Adams 1996) (Eskling et al. 1997). The DES reflects the activity of this excess energy dissipation cycle where zeaxanthin is the most efficient xanthophyll. Higher values correspond to a higher activity of this cycle trying to convert the epoxides into zeaxanthin (Duarte et al. 2015).

3.5. Photobiology

Pulse amplitude modulated (PAM) fluorometry was used to monitor photosynthetic activity (Monitoring Pen MP 100, Photon Systems Instruments, Czech Republic). The chlorophyll fluorescence measurement parameter used was the OJIP (Chlorophyll Fluorescence Induction Kinetics), with a protocol that captures rapid fluorescence transient (which occurs during exposure to high irradiance) thus providing information about the functioning of photosystem II (PSII) (Strasser et al. 2004). Measurements were performed with a super pulse (default intensity of the saturating light pulse - 80%) which induces a maximum chlorophyll fluorescence intensity (F_m).

Prior to OJIP measurements, all experimental replicates were dark adapted for 20 minutes, to allow that all photosynthetic reaction centers and the electron transport chain (ETC) were in a relaxed/open state ($=F_0$). The experimental replicates were allocated in a dark room and immersed in a water bath which maintained the experimental temperature conditions being set for that specific day. OJIP measurements ($n=10$ per replicate) were performed in submerged seagrass leaves, during specific time points of the experimental exposure (*i.e.* T1, T2, T3 and T4). The FluorPen 1.1 software (Photon Systems Instruments, Czech Republic) was used for data collection and processing.

The electron transport rate (ETR), a light-adapted parameter that is directly related to PSII was calculated according to:

$$ETR = F_v/F_m \times PAR \times 0.5 \times AF$$

where F_v/F_m represents the maximum quantum yield of PSII which provides information on the functioning of the PSII and therefore it is used for detecting stress from adverse conditions in plants, and 0.5 compensates for irradiance being split between the two photosystems, meaning that assumes equal distribution of absorbed photons between photosystem I (PSI) and PSII. The leaf absorption factor (AF) used was 0.84.

3.6. Macrofauna Diversity

Seagrass patches and its sediment were sieved at the end of the recovery period (T4) with a 2.00 mm-mesh sieve. All organisms were separated into macrofaunal classes (*i.e.* Bivalvia, Gastropoda and Malacostraca). Subsequently, each organism was sampled and preserved (70% alcohol), for posterior identification. Species identification was performed under stereomicroscope observations (SMZ-171, Motic, China) and based on Hayward and Ryland (Hayward & Ryland 2017). Incomplete animals which had the anterior end missing were not recorded as individuals. Based on species and specimen abundance values, several diversity indices were calculated and applied to describe biodiversity patterns and assess changes in the structure of macrofaunal assemblages associated with seagrasses. Nearly all diversity and evenness indices are based on the relative abundance of species (p_i) in which $p_i = n_i / N$, where n_i is the abundance of the i -th species in the sample and N is the total number of individuals in the community. The following biodiversity indices were assessed:

Shannon-Wiener diversity index (H) in order to measure the uncertainty in the identification of a randomly chosen individual on the assemblage (Shannon & Weaver 1949):

$$H = - \sum_i^s p_i \ln p_i$$

Simpson's diversity index (λ) to assess the probability that two randomly chosen individuals belong to different species (Simpson 1949):

$$\lambda = 1 - \sum_i^s \frac{n_i(n_i-1)}{(N-1)}$$

Margalef's diversity index (D) to determine species richness, which gives the number of species present for a given number of individuals (Margalef 1958):

$$D = \frac{S-1}{\ln N}$$

where S is the total number of species and N the total number of individuals (organisms) in the community.

Pielou's evenness index (J) which expresses how evenly the individuals are distributed among the different species (Pielou 1966):

$$J = \frac{H}{H_{\max}}$$

where $H_{\max}$ is the maximum possible value of Shannon-Wiener diversity for a given number of individuals and species ($H_{\max} = \log S$, where S is the total number of species in the sample).

4. Data Analysis

All statistical analyses and graphs were performed in RStudio Software (Version 4.0.3 - © 2020 RStudio, Inc.). Prior to statistical analyses, data was tested for homogeneity of variance, normality of distribution and a regression analysis was used to identify influential data points (outliers) that may negatively affect the regression model. Shoot density, photobiological (ETR and F_v/F_m) and biochemical (pigments) variables and macrofaunal diversity were statistically treated with parametric generalized linear models (GLM) where Gaussian was selected to determine the probability distribution of differences between the three treatment groups. Significance of GLM was assumed when the probability was lower than 0.05 and had been confirmed using Type-II test sum of squares using “Anova” function from the package “Car” (Fox & Weisberg 2019) and “emmeans” function from the package “emmeans” (Russell et al. 2021). Data exploration and model validation used the HighstatLibV10 R library from Highland Statistics (Zuur et al. 2009) following (Zuur et al. 2010). Experimental replicates were included as a random effect.

5. Results

5.1. Shoot density and photobiological response

At the beginning of the experiment trials (T0), seagrass shoot density was 0.60 ± 0.11 (mean $\pm$ SD) shoots cm^{-2} , and at the end of the recovery period (T4), it varied between: i) 0.49 ± 0.18 shoots cm^{-2} (control conditions), ii) 0.60 ± 0.28 shoots cm^{-2} (MHW category III) and iii) 0.60 ± 0.19 shoots cm^{-2} (MHW category IV). No significant thermal-induced differences in shoot density ($p\text{-value} > 0.05$; GLM; Gaussian family; Table 5.1.1) were found among experimental treatments (*i.e.* control, MHW category III and MHW category IV) (Figure 5.1.1).

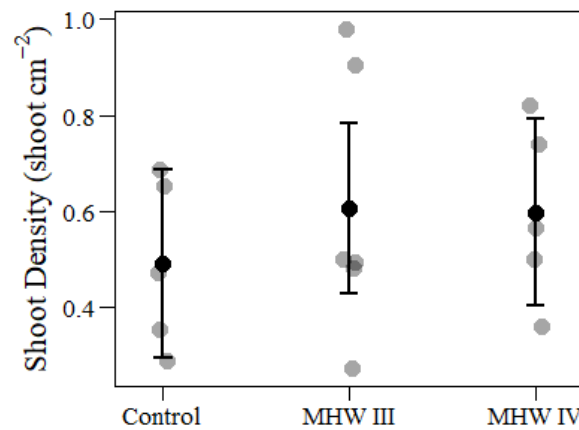


Figure 5.1.1 *Zostera noltei* shoot density (sampling point T4) after the exposure to different experimental conditions (*i.e.* control, MHW category III and MHW category IV). Values represent mean $\pm$ SD ($n = 5$ for control; $n = 6$ for MHW category III; $n = 5$ for MHW category IV) in number of shoots cm^{-2} . Abbreviations: MHW III: MHW category III; MHW IV: MHW category IV.

Table 5.1.1 Analysis of deviance table (Type II tests) for the analysis of *Zostera noltei* shoot density. Statistical significance at $p\text{-value} < 0.05$.

Model: GLM (Gaussian)		Response variable: Shoot Density	
Model term(s): MHW			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	0.8617	0.65

No significant changes on ETR ($p\text{-value} > 0.05$; GLM; Gaussian family; Table 5.1.2) and F_v/F_m ($p\text{-value} > 0.05$; GLM; Gaussian family; Table 5.1.3) were observed between the different treatments and within all sampling points (T1 -T4) (Figure 5.1.2).

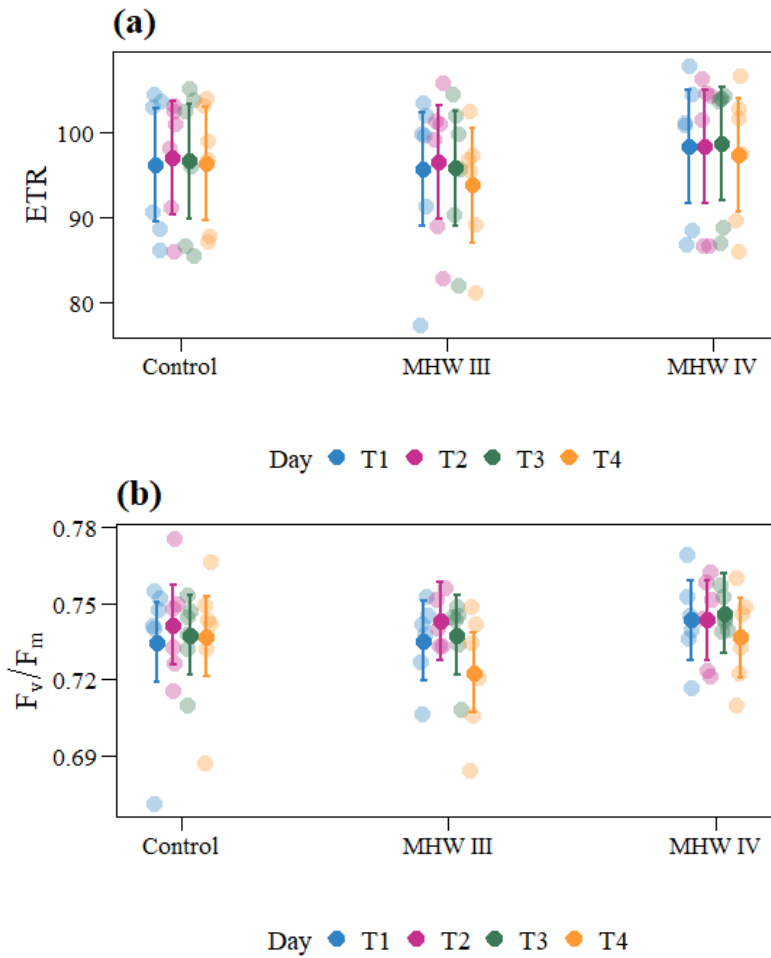


Figure 5.1.2 Electron transport rate (ETR) and maximum PSII quantum yield (F_v/F_m) values measured in *Zostera noltei* leaves, exposed to different experimental conditions (i.e. control, MHW category III and MHW category IV) and at different sampling points (i.e. T1-T4). Values represent mean $\pm$ SD (n = 6 for control; n = 6 for MHW category III; n = 6 for MHW IV) for (a) ETR and (b) F_v/F_m . Abbreviations: MHW III: MHW category III; MHW IV: MHW category IV.

Table 5.1.2 Analysis of deviance table (Type II tests) for the analysis of Electron transport rate (ETR) values in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: ETR	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	0.3471	0.8407
Day	3	4.6213	0.2017
MHW x Day	6	2.4015	0.8793

Table 5.1.3 Analysis of deviance table (Type II tests) for the analysis of maximum PSII quantum yield (F_v/F_m) values in *Zostera noltei* leaves. Statistical significance at p -value<0.05.

Model: GLM (Gaussian)		Response variable: F_v/F_m	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	0.9928	0.6087
Day	3	4.3015	0.2307
MHW x Day	6	2.4290	0.8763

5.2. Leaf pigment composition

No significant differences were observed in chlorophyll *a* (p -value>0.05; GLM; Gaussian family; Table 5.2.1; Figure 5.2.1a), chlorophyll *b* (p -value>0.05; GLM; Gaussian family; Table 5.2.2; Figure 5.2.1b), total chlorophyll (p -value>0.05; GLM; Gaussian family; Table 5.2.3; Figure 5.2.1c) and pheophytin *b* (p -value>0.05; GLM; Gaussian family; Table 5.2.5; Figure 5.2.1e) concentrations in *Z. noltei* leaves, between the different experimental treatments (*i.e.* control, MHW category III and MHW category IV) and within the sampling points (*i.e.* T3 and T4). In pheophytin *a* concentration values, a significant effect was detected for the MHW factor (p -value<0.05; GLM; Gaussian family; Table 5.2.4; Figure 5.2.1d). However, after a multi-comparison analysis, the factor turned out not to be statistically significant (Supplementary Table 2).

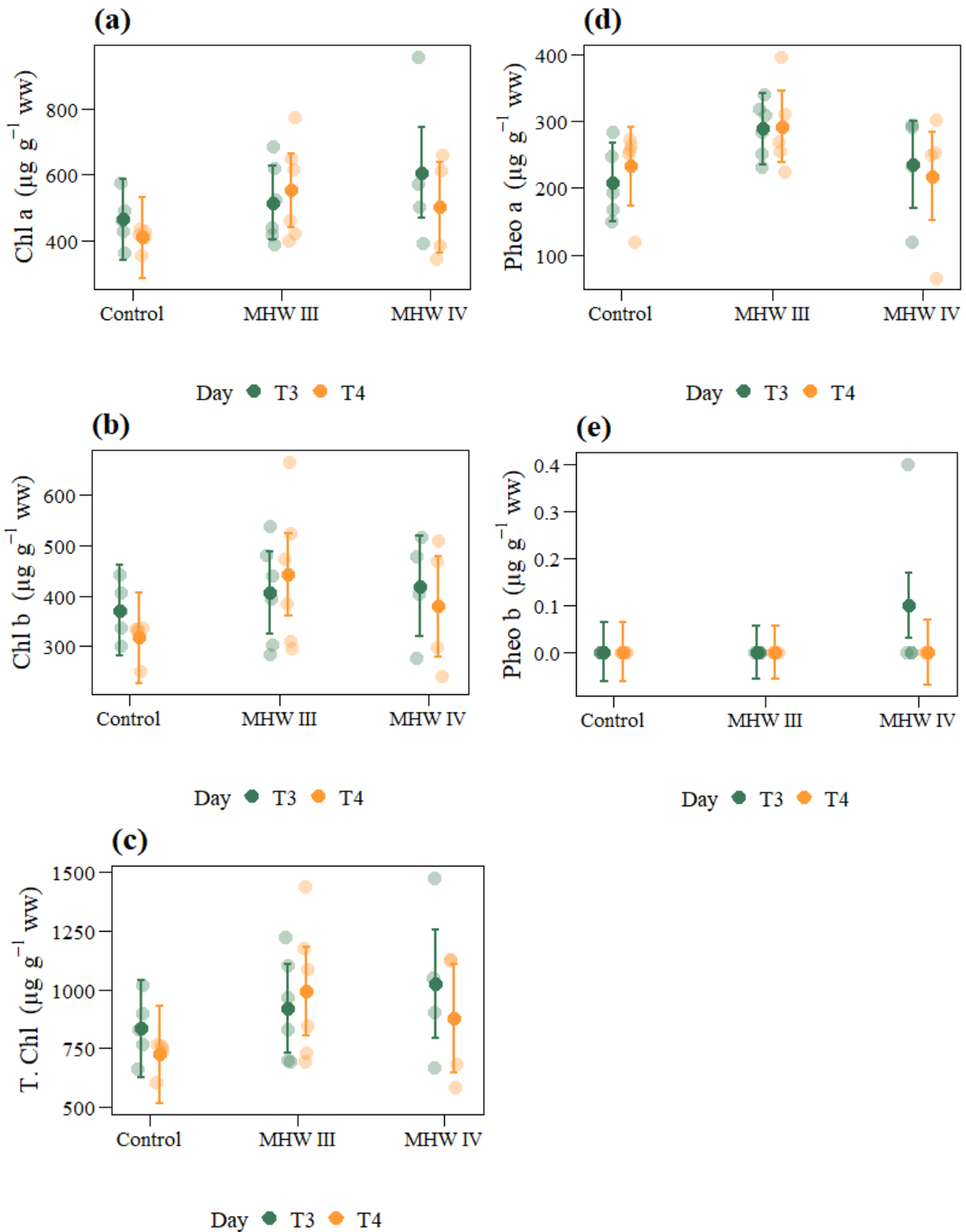


Figure 5.2.1 Chlorophyll (*a*, *b* and Total) and Pheophytin (*a* and *b*) concentration in *Zostera noltei* leaves, exposed to different experimental conditions (*i.e.* control, MHW category III and MHW category IV) and at different sampling points (*i.e.* T3 and T4). Values represent mean $\pm$ SD ($n = 10$ for control; $n = 12$ for MHW category III; $n = 8$ for MHW category IV) for: a) Chlorophyll *a*; b) Chlorophyll *b*; c) Total Chlorophyll; d) Pheophytin *a* and e) Pheophytin *b* concentrations in $\mu\text{g g}^{-1} \text{ ww}$. Abbreviations: Chl *a*: Chlorophyll *a*; Chl *b*: Chlorophyll *b*; T. Chl: Total Chlorophyll; Pheo *a*: Pheophytin *a*; Pheo *b*: Pheophytin *b*; MHW III: MHW category III; MHW IV: MHW category IV.

Table 5.2.1 Analysis of deviance table (Type II tests) for the analysis of Chlorophyll *a* concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

<i>Model: GLM (Gaussian)</i>		<i>Response variable: Chlorophyll a</i>	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	3.6190	0.1637
Day	1	0.3831	0.5359
MHW x Day	2	1.5168	0.4684

Table 5.2.2 Analysis of deviance table (Type II tests) for the analysis of Chlorophyll *b* concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

<i>Model: GLM (Gaussian)</i>		<i>Response variable: Chlorophyll b</i>	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	3.4304	0.1799
Day	1	0.1529	0.6958
MHW x Day	2	1.2296	0.5407

Table 5.2.3 Analysis of deviance table (Type II tests) for the analysis of Total Chlorophyll concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

<i>Model: GLM (Gaussian)</i>		<i>Response variable: Total Chlorophyll</i>	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	3.6006	0.1652
Day	1	0.2812	0.5959
MHW x Day	2	1.3693	0.5043

Table 5.2.4 Analysis of deviance table (Type II tests) for the analysis of Pheophytin *a* concentrations in *Zostera noltei* leaves. Values of p-value<0.05 are shown in bold.

<i>Model: GLM (Gaussian)</i>		<i>Response variable: Pheophytin a</i>	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	7.1831	0.02756
Day	1	0.0362	0.84902
MHW x Day	2	0.4269	0.80780

Table 5.2.5 Analysis of deviance table (Type II tests) for the analysis of Pheophytin *b* concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: Pheophytin b	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	2.9333	0.2307
Day	1	1.0667	0.3017
MHW x Day	2	2.9333	0.2307

Alongside, no significant effects were observed in lutein (p-value>0.05; GLM; Gaussian family; Table 5.2.7; Figure 5.2.2b), violaxanthin (p-value>0.05; GLM; Gaussian family; Table 5.2.8; Figure 5.2.2c), β -Carotene (p-value>0.05; GLM; Gaussian family; Table 5.2.9; Figure 5.2.2d), total auroxanthin (*i.e.* aurochrome concentration + auroxanthin concentration) (p-value>0.05; GLM; Gaussian family; Table 5.2.10; Figure 5.2.2e), and zeaxanthin (p-value>0.05; GLM; Gaussian family; Table 5.2.11; Figure 5.2.2f) concentrations between the different experimental treatments. A similar trend was observed for carotene/chlorophyll ratio (p-value>0.05; GLM; Gaussian family; Table 5.2.13; Figure 5.2.3b) and DES (p-value>0.05; GLM; Gaussian family; Table 5.2.14; Figure 5.2.3c) values. In antheraxanthin and total carotenoid concentration values, a significant effect was detected for the MHW factor (p-value<0.05; GLM; Gaussian family; Table 5.2.6 and 5.2.12 respectively; Figure 5.2.2a and 5.2.3a). However, after a multi-comparison analysis, the factor turned out not to be statistically significant (Supplementary Table 3 and 4).

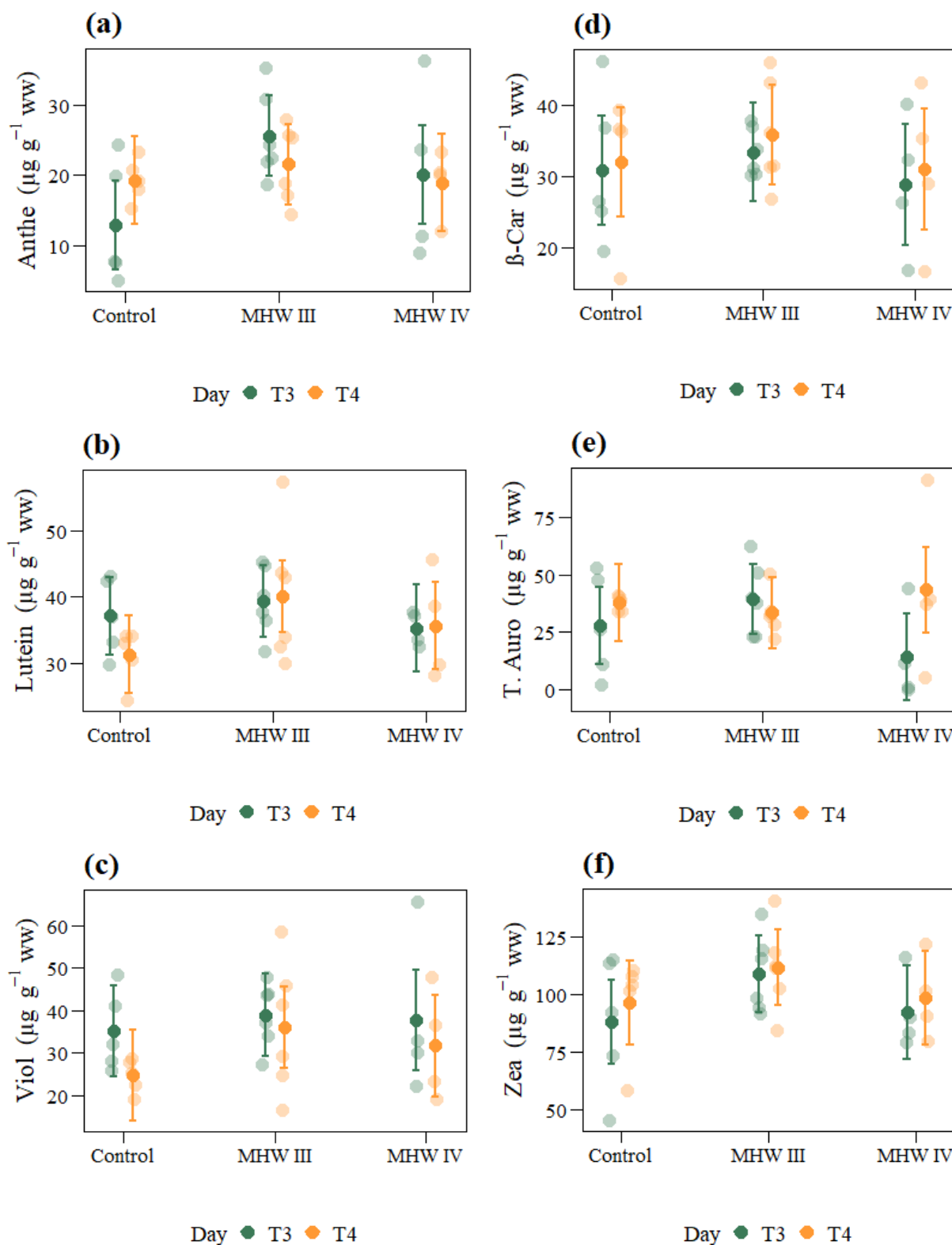


Figure 5.2.2. Antheraxanthin, Lutein, Violaxanthin, β -Carotene, Total Auroxanthin and Zeaxanthin concentrations in *Zostera noltei* leaves, exposed to different experimental conditions (*i.e.* control, MHW category III and MHW category IV) and at different sampling points (*i.e.* T3 and T4). Values represent mean $\pm$ SD ($n = 5$ for control; $n = 6$ for MHW category III; $n = 4$ for MHW category IV) for: a) Antheraxanthin; b) Lutein; c) Violaxanthin; d) β -Carotene; e) Total Auroxanthin and f) Zeaxanthin concentrations in $\mu\text{g g}^{-1} \text{ ww}$. Abbreviations: Anth: Antheraxanthin; Viol: Violaxanthin; β -car: β -carotene; T. Auro: Total Auroxanthin; Zea: Zeaxanthin; MHW III: MHW category III; MHW IV: MHW category IV.

Table 5.2.6 Analysis of deviance table (Type II tests) for the analysis of Antheraxanthin concentrations in *Zostera noltei* leaves. Values of p-value<0.05 are shown in bold.

Model: GLM (Gaussian)		Response variable: Antheraxanthin	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	6.0280	0.04909
Day	1	0.0071	0.093274
MHW x Day	2	2.9917	0.22405

Table 5.2.7 Analysis of deviance table (Type II tests) for the analysis of Lutein concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

<i>Model: GLM (Gaussian)</i>		<i>Response variable: Lutein</i>	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	4.1287	0.1269
Day	1	0.4178	0.5180
MHW x Day	2	1.5509	0.4605

Table 5.2.8 Analysis of deviance table (Type II tests) for the analysis of Violaxanthin concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: Violaxanthin	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	1.9364	0.3798
Day	1	2.2309	0.1353
MHW x Day	2	0.5807	0.7480

Table 5.2.9 Analysis of deviance table (Type II tests) for the analysis of β -Carotene concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: β -Carotene	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	1.5337	0.4645
Day	1	0.3710	0.5425
MHW x Day	2	0.0312	0.9845

Table 5.2.10 Analysis of deviance table (Type II tests) for the analysis of Total Auroxanthin concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

<i>Model: GLM (Gaussian)</i>		<i>Response variable: Total Auroxanthin</i>	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	0.7950	0.6720
Day	1	1.4986	0.2209
MHW x Day	2	4.1418	0.1261

Table 5.2.11 Analysis of deviance table (Type II tests) for the analysis of Zeaxanthin concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: Zeaxanthin	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	4.7577	0.09266
Day	1	0.5487	0.45886
MHW x Day	2	0.1063	0.94826

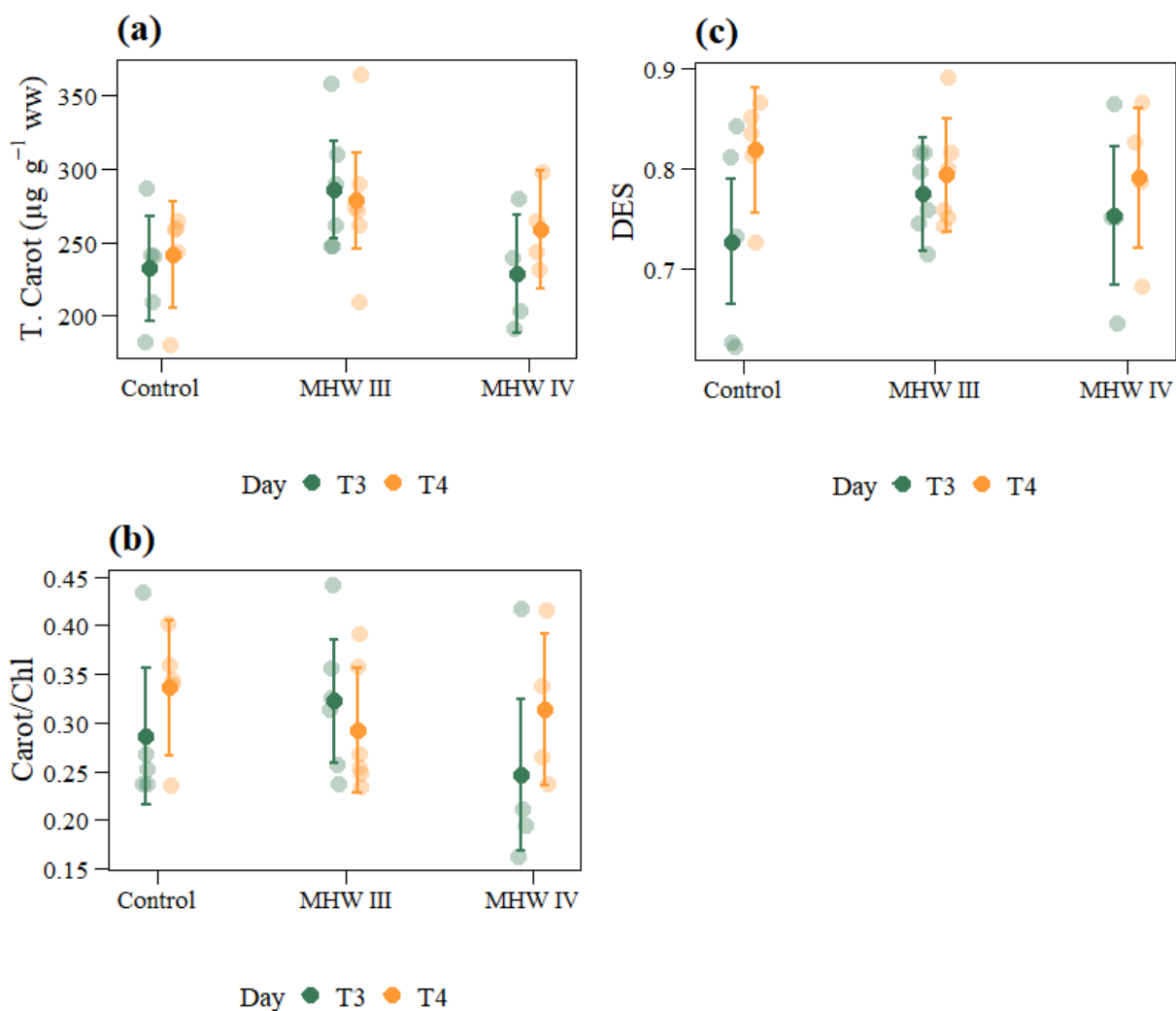


Figure 5.2.3 Total Carotenoids concentration, Carotene/Chlorophyll ratio and DES values in *Zostera noltei* leaves, exposed to different experimental conditions (*i.e.* control, MHW category III and MHW category IV) and at different sampling points (*i.e.* T3 and T4). Values represent mean $\pm$ SD ($n = 5$ for control; $n = 6$ for MHW category III; $n = 4$ for MHW category IV) for: a) Total Carotenoids content ($\mu\text{g g}^{-1} \text{ ww}$); b) Carotene/Chlorophyll ratio and c) De-Epoxidation State (DES). Abbreviations: T. Carot: Total Carotenoids; Carot/Chl: carotenoid/chlorophyll ratio; DES: De-epoxidation state; MHW III: MHW category III; MHW IV: MHW category IV.

Table 5.2.12 Analysis of deviance table (Type II tests) for the analysis of Total Carotenoids concentrations in *Zostera noltei* leaves. Values of p-value < 0.05 are shown in bold.

Model: GLM (Gaussian)		Response variable: Total Carotenoids	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	7.7506	0.02075
Day	1	0.3171	0.57334
MHW x Day	2	1.0312	0.59715

Table 5.2.13 Analysis of deviance table (Type II tests) for the analysis of Carotene/Chlorophyll ratio in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: Carotene/Chlorophyll ratio	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	0.7922	0.6729
Day	1	0.6060	0.4363
MHW x Day	2	2.2640	0.3224

Table 5.2.14 Analysis of deviance table (Type II tests) for the analysis of DES in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: DES	
Model term(s): MHW * Day			
Random effects: Experimental replicate			
Factor:	df	χ^2	P
MHW	2	0.1817	0.9132
Day	1	3.7408	0.0531
MHW x Day	2	1.6285	0.4430

5.3. Macrofauna Diversity

A total of 9 species, comprising a total of 247 specimens, belonging to the classes Bivalvia, Gastropoda and Malacostraca, were identified. Gastropods were dominant in terms of species number, number of individuals and presence throughout all experimental treatments. *Peringia ulvae* was the most abundant species across all treatments (Supplementary Table 5). The Shannon-Wiener diversity index (H) varied between 0.88 (control) and 1.10 (MHW category III) (Figure 5.3.1a). The Simpson's diversity index (λ) achieved its maximum value under MHW category III (0.59) and its lowest value under control conditions (0.53) (Figure 5.3.1b). Margalef's diversity index (D) values varied between 0.96 and 1.37, under control and MHW category III conditions respectively (Figure 5.3.1c). Regarding Pielou's evenness index (J), values varied between 0.69 (MHW category IV) and 0.74 (control) (Figure 5.3.1d). None of these indexes revealed significant changes among treatments (p-value>0.05; GLM; Gaussian family; Tables 5.3.1, 5.3.2, 5.3.3 and 5.3.4).

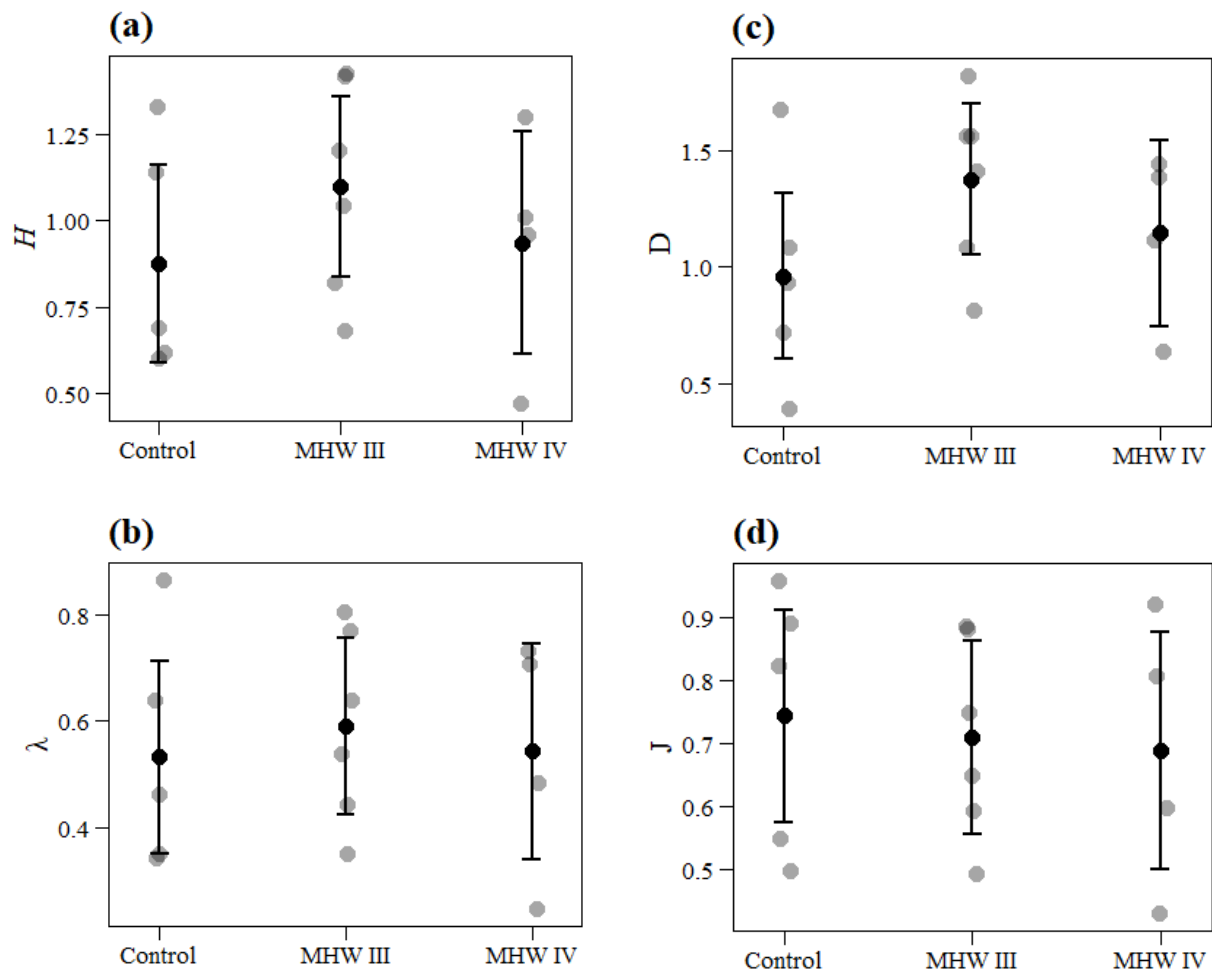


Figure 5.3.1 Shannon-Wiener, Simpson's, Margalef's and Pielou's evenness indexes for *Zostera noltei* associated macrofauna (sampling point T4) after the exposure to different experimental conditions (*i.e.* control, MHW category III and MHW category IV). Values represent mean $\pm$ SD ($n = 5$ for control; $n = 6$ for MHW category III; $n = 4$ for MHW category IV) for: a) Shannon-Wiener diversity index (H); b) Simpson's diversity index (λ); c) Margalef's diversity index (D) and d) Pielou's evenness index (J). Abbreviations: MHW III: MHW category III; MHW category IV: MHW category IV.

Table 5.3.1 Analysis of deviance table (Type II tests) for the analysis of Shannon-Wiener diversity index for *Zostera noltei* associated macrofauna. Statistical significance at p -value < 0.05 .

Model: GLM (Gaussian)		Response variable: Shannon-Wiener diversity index	
Model term(s): MHW			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	1.3765	0.5025

Table 5.3.2 Analysis of deviance table (Type II tests) for the analysis of Simpson's diversity index for *Zostera noltei* associated macrofauna. Statistical significance at p-value<0.05.

<i>Model: GLM (Gaussian)</i>		<i>Response variable: Simpson's diversity index</i>	
Model term(s): MHW			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	0.25008	0.8825

Table 5.3.3 Analysis of deviance table (Type II tests) for the analysis of Margalef's diversity index for *Zostera noltei* associated macrofauna. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: Margalef's diversity index		
Model term(s): MHW				
Random effects: Experimental replicate				
Factor:		df	χ^2	p
MHW		2	2.8683	0.2383

Table 5.3.4 Analysis of deviance table (Type II tests) for the analysis of Pielou's evenness index for *Zostera noltei* associated macrofauna. Statistical significance at p-value<0.05.

Model: GLM (Gaussian)		Response variable: Pielou's evenness index	
Model term(s): MHW			
Random effects: Experimental replicate			
Factor:	df	χ^2	p
MHW	2	0.19477	0.9072

6. Discussion

6.1. Temperature effects on *Z. noltei*

Marine heat waves can pose deleterious effects on seagrass meadows by directly affecting their structure and functioning (Arias-Ortiz et al. 2018). Moreover, the ability of seagrasses to endure temperature changes is dependent on each species thermal threshold (Short & Neckles 1999). Considering IPCC projections, upper temperature thresholds tolerance of seagrasses species may be exceeded within the next decades, which in turn may potentiate the occurrence of mass shoot mortality, damage to the plant's physiological processes and ultimately their survival (Marbà & Duarte 2010). Yet here *Z. noltei* shoot density was not affected by experimentally simulated MHW conditions. These results conflict with previous reports that identified ocean warming as a potential stressor which had a direct impact in seagrasses shoot mortality/loss. In fact, a MHW in 2003 followed by another in 2006 in the Mediterranean sea had such a considerable impact in *P. oceanica* shoot mortality that even with an increase in the recruitment rate, the meadow was unable to recover (Marbà & Duarte 2010). In another study, *Z. noltei* sensitivity to a small temperature increase was observed, i.e. an increase of 2°C (from 37°C to 39°C) led to a shoot mortality increase of more than 95% (Massa et al. 2009). Repolho et al. (2017) also described a significant decrease in shoot density of *Z. noltei* when exposed to increased warming conditions. However, these authors performed an exposure of *Z. noltei* to a constant and increased temperature condition (22 °C) throughout 1 month, which is not identical to the present MHW experimental design. These present results may be explained by the fact that *Z. noltei* was harvested in an estuary area which is subject to tidal regimes and high temperature fluctuations (Neto et al. 2019), which to some extent can confer a higher acclimation and/or adaption capacity to withstand short-term temperature extremes.

As for submerged aquatic plants (Hough & Wetzel 1977), seagrasses were previously categorized as mostly C3 plants (Andrews & Abel 1979). However, some of these plants have also shown a C4 pathway and others are C3-C4 intermediates (Beer et al. 1980) making it a subject not very consensual in the scientific community. F_v/F_m measurements on dark-adapted leaves is commonly used to investigate the onset and recovery from stress situations (Murchie & Lawson 2013) as a proxy to measure the rate of linear electron transport, suggesting the photosynthetic capacity of the plant (Ashraf & Harris 2013). A study conducted by Björkman and Demmig (1987) reported that all analyzed species (C3, C4 and CAM) had F_v/F_m values between 0.78 and 0.86 for healthy plants. In another study, the authors recommend that in fluorescence analysis, non-stressed plants should have maximum PSII quantum yield values of 0.81-0.83 (Murchie & Lawson 2013). Values lower than these could suggest a limited damage in the PSII reaction centers, inducing a malfunctioning of the photosynthetic apparatus and/or photoinhibition in response to temperature (Maxwell & Johnson 2000, Ashraf & Harris 2013).

In order to assess *Z. noltei* stress level and to check if PSII photochemistry potential remained intact during and after MHWs conditions, chlorophyll fluorescence measurements of the maximal quantum yield of PSII were performed and ETR was calculated.

In the present dissertation, the observed F_v/F_m and ETR values for *Z. noltei* exposed to control and temperature increased conditions indicated that the temperature profile experienced by the plant under MHW category IV conditions still allows the functioning of the plant photosynthetic machinery. Moreover, these results could indicate, once more, the ability of this species to be already adapted to fluctuating temperature (e.g. extreme) conditions. Previous studies have observed a drastic F_v/F_m ratio and ETR decline in several tropical species exposed to increased temperatures, due to photochemical efficiency inhibition and damage in photosynthetic pathways (Campbell et al. 2006, George et al. 2018). However tropical seagrass species are generally found in environments where water temperatures reach

their upper limit of thermal tolerance, meaning they are only able to tolerate small changes in temperature, making them more vulnerable than temperate seagrasses (Ralph 1998, Campbell et al. 2006). For the same species in our study, *Z. noltei* collected in a nearby area close to sampling location, had a significant decrease in F_v/F_m values and ETR capacity under temperatures of 22°C (with 18°C as control) (Repolho et al. 2017).

Yet, the present results suggest that the MHW scenarios tested may not pose a significant negative impact in *Z. noltei* photosynthetic apparatus.

Photosynthetic pigment analysis is also used as a proxy to determine health status of the plant, besides its overall photosynthetic capacity (Liu et al. 2011). Casazza and Mazzella (2002) showed that these marine angiosperms share a similar pigment profile to land plants, including chlorophylls (chlorophyll *a* and *b*) with direct functions in photosynthesis and carotenoids (xanthophylls and carotenes) as accessory pigments to protect chlorophyll from photooxidation (Fyfe 2003). Since temperature has a direct effect on the enzymatic activity and chlorophyll biosynthesis, it would be expectable that MHW conditions could influence pigment concentrations (Niu et al. 2012). Chlorophylls have a central and fundamental role during photosynthesis because they are the specialized light-absorbing pigments. Thus, this primary pigment has been used as an indicator of plant status (Stockburger & Mitchell 1999) and, especially in seagrasses under high temperatures, chlorophyll content is used to identify chloroplast and photosynthesis malfunctions (Purnama et al. 2018). Yet, in the present dissertation no significant differences in chlorophyll *a* and *b*, and total chlorophyll concentrations were observed. These findings contradict previous studies that recorded a significant decrease in chlorophyll content within *Thalassia hemprichii* leaves, in response to high temperatures conditions (Purnama et al. 2018). Also, a decrease in chlorophyll content was previously reported in *Z. noltei* seagrasses under heat stress for 30 consecutive days (Repolho et al. 2017).

The xanthophyll cycle performance in excess energy dissipation was assessed through the DES where violaxanthin is photoconverted in zeaxanthin through antheraxanthin (Adams & Demmig-Adams 1992, Ashraf & Harris 2013). An increase in DES suggests an activation of the xanthophyll cycle (Duarte et al. 2012). Another mechanism for photoprotection is through non-photochemical quenching (NPQ) by auroxanthin (Ruban et al. 1998). Auroxanthin is a zeaxanthin analogue that has previously been shown to behave as a carotenoid capable of stimulating quenching (Ruban et al. 1998, Wentworth et al. 2000). Lutein is another carotene structurally similar to zeaxanthin and it is also one of the constituent pigments of light-harvesting complexes (LHCs). It possesses a key role in NPQ and therefore, like zeaxanthin and antheraxanthin, it also has the function of protecting plants from photo-oxidative stress (Thayer & Björkman 1990, Niyogi et al. 1997). Alongside, β -carotene also plays an important role in photosynthesis regulation and dissipation of excess energy (Kamiya & Shen 2003) by quenching the triplet states of chlorophyll (Repolho et al. 2017). Yet, in the present work, antheraxanthin, lutein, violaxanthin, β -Carotene, total auroxanthin, zeaxanthin and total carotenoid concentrations, carotene/chlorophyll ratio and DES levels also did not change among the different thermal treatments.

6.2. Temperature effects on seagrass-associated macrofauna

Previous studies have analyzed the impacts that MHWs have on the structure and composition of macroinvertebrate communities. These extreme heat events could pose a decline in species richness and in the abundance of individuals, as well as a change in their distribution (Sorte et al. 2010, Smale et al. 2017). Thereby, severe effects are expected in organisms subjected to sudden thermal stress as MHWs that can ultimately lead to mortality, especially for the ones that are not highly mobile or sessile such as corals, polychaetes, gastropods, bivalves or sea urchins (Leung et al. 2017). The effects of MHWs on macrofauna biodiversity can be assessed based on the duration and magnitude of the extreme

event and also on the adaptative capacity and tolerance that organisms may have to cope with thermal stress (Frölicher & Laufkötter 2018, Leung et al. 2019). For instance, a study conducted by Hemraj et al. (2020) showed how two species of gastropods with different distributions (*i.e.* *Trochus sacellum* as a tropical species and *Astrarium haematragum* as a temperate species) had similar mortality rates when exposed to MHWs. Although both species experienced 100% mortality under an extreme MHW, the tropical species reached total mortality faster than the temperate one. This suggests that tropical species have a narrower thermal tolerance range and therefore a limited acclimation capacity when compared to temperate species that can adapt to larger environmental variations (Rosa et al. 2014a b). Comparable effects have also been observed in the temperate gastropod *Thalotia conica* under prolonged exposure (*i.e.* 8 weeks) to MHWs, with significant decreases in survival (Leung et al. 2017).

In the present study, all the diversity indicators, namely the Shannon-Wiener (H), Simpson's (λ) and Margalef's (D) diversity indexes, and Pielou's evenness index (J), did not change among treatments. It is worth noting that all species identified are known to tolerate and maintain their performance when exposed to high temperatures (Evans 1948, Zaouali 1974, Lassen & Kristesen 1978, Budd 2007, Gardeström et al. 2007, Madeira et al. 2012). Yet, a recent experience found out that four different estuarine bivalve species (*Cerastoderma edule*, *Ruditapes philippinarum*, *Venerupis corrugate* and *Ruditapes decussatus*) experienced an increase in mortality when exposed to MHWs (Domínguez et al. 2021). In another study, the Mediterranean mussel *Mytilus galloprovincialis* acclimated at 18°C started to die at 26°C. A significant increase in mortality occurred after 5 days exposure when water temperature reached 30°C due to a diminished capacity to assimilate food (Anestis et al. 2007).

In the present dissertation, the macrofaunal community of the Sado estuary *Z. noltei* meadow had low Shannon-Wiener diversity index when compared to other seagrass meadows (Boström & Bonsdorff 1997, McCloskey & Unsworth 2015). The low value indicates that our study site was dominant by only a few species. Margalef's diversity index also had low values when compared to other seagrass meadows (Colletta et al. 1984) meaning that there is low species diversity. Regarding Pielou's evenness index and the Simpson's diversity index, both obtained values lower than the ones in a seagrass bed in the Philippines (Mendoza et al. 2019). This suggests that there is low diversity in the community due to a low number of species. Because the Sado estuary has larger temperature fluctuations during the day, macrofauna species might be adapted to higher temperatures such as the ones observed during the summer period (Neto et al. 2019). Furthermore, the study area of the estuary near the Eurominas region was previously identified as one of the regions with the lowest species richness, abundance and diversity values suggesting that this area is one of the most disturbed (Rodrigues 1992, Carvalho et al. 2000). This might justify the lack of significant differences and the low biodiversity numbers in all experimental treatments.

7. Conclusions

Extreme weather events are predicted to occur more frequently in the future, diminishing the resilience of foundation species, with special emphasis to those that are already close to their thermal threshold. This events such as marine heat waves will dictate the response of seagrass ecosystems to climate change (Fraser et al. 2014). If the trend continues, seagrass meadows will continue to disappear worldwide at an alarming rate due to increases in natural and human-induced disturbances (Orth et al. 2006, Waycott et al. 2009). The results of this dissertation clearly showed that *Z. noltei* and associated macrofauna may be resilient to short-term extreme MHW conditions. Yet, under long-term/chronic warming conditions expected until the end of the century such plasticity and adaptive capacity may not suffice (Repolho et al. 2017). The study of the impact of climate change on seagrass meadows remains essential especially because seagrass meadows are among the most productive ecosystems on the planet (Beck et al. 2001, Green & Short 2003). Therefore, there is an urgent need to conserve seagrass ecosystems wherever possible and predict species resilience to the IPCC possible future climate scenarios.

8. References ¹

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9. Supplementary data

Supplementary Table 1 Seawater physicochemical parameters in all experimental setups. Temperature, pH, dissolved oxygen and salinity were measured daily and averaged per experimental replicate aquarium over the whole experimental period.

Period	Day	Treatment	Expected Temperature (°C)	Measured Temperature (°C)	pH	O2 (mg L ⁻¹)	O2 (%)	Salinity (ppt)
Acclimatation	11/06/2020	Control	18.0	18.1	8.176	9.52	124.0	35.2
		MHW category III	18.0	18.1	8.217	10.46	136.2	35.2
		MHW category IV	18.0	17.9	8.139	8.88	115.4	35.6
	12/06/2020	Control	18.0	18.4	8.124	8.62	113.1	35.4
		MHW category III	18.0	18.4	8.153	9.06	118.7	35.2
		MHW category IV	18.0	18.4	8.108	8.31	108.8	35.0
	13/06/2020	Control	18.0	17.9	8.094	9.15	118.5	35.0
		MHW category III	18.0	18.0	8.166	9.63	125.8	36.0
		MHW category IV	18.0	17.8	8.077	8.79	113.7	35.0
	14/06/2020	Control	18.0	18.2	8.158	10.10	131.6	35.0
		MHW category III	18.0	18.3	8.181	10.17	132.8	35.0
		MHW category IV	18.0	18.1	8.119	9.68	125.8	35.0
	15/06/2020	Control	18.0	18.7	8.140	9.36	123.3	35.0
		MHW category III	18.0	18.6	8.172	10.15	133.4	35.0
		MHW category IV	18.0	18.5	8.129	9.49	124.6	35.0
Temperature ramp	16/06/2020	Control	17.9	17.9	8.031	8.11	105.7	36.0
		MHW category III	17.9	18.1	8.055	8.24	107.6	35.8
		MHW category IV	17.9	18.0	8.010	8.05	105.1	35.8
	17/06/2020	Control	17.9	18.2	8.080	8.93	116.5	35.0
		MHW category III	18.3	18.3	8.078	8.58	112.1	35.0
		MHW category IV	18.4	18.6	8.050	8.59	112.9	35.0
	18/06/2020	Control	17.9	18.1	8.065	8.22	107.5	36.0
		MHW category III	18.7	18.4	8.074	8.22	108.1	36.0
		MHW category IV	19.0	18.9	8.056	8.15	108.1	35.6
	19/06/2020	Control	18.00	18.4	8.145	8.77	114.7	35.0
		MHW category III	19.1	19.1	8.167	8.42	111.6	35.0
		MHW category IV	19.5	19.4	8.135	8.40	112.1	35.0
	20/06/2020	Control	18.0	18.2	8.080	8.60	112.0	35.0
		MHW category III	19.5	19.3	8.125	8.33	110.9	35.0
		MHW category IV	20.0	19.8	8.102	8.39	112.7	35.0
	21/06/2020	Control	18.0	18.3	8.164	8.49	111.6	36.0
		MHW category III	19.9	19.6	8.180	8.09	108.6	35.5
		MHW category IV	20.6	20.3	8.165	8.17	111.6	36.0
	22/06/2020	Control	18.1	18.3	8.134	8.80	114.9	35.0
		MHW category III	20.4	20.4	8.172	8.61	117.1	35.0
		MHW category IV	21.1	20.9	8.168	8.52	117.0	35.0
	23/06/2020	Control	18.1	18.4	8.171	8.66	114.0	36.0
		MHW category III	20.8	20.7	8.213	8.21	112.9	36.0

Temperature plateau	24/06/2020	MHW category IV	21.6	21.2	8.222	8.21	114.1	36.0
		Control	18.1	18.2	8.139	8.51	110.9	35.0
		MHW category III	21.2	21.0	8.175	8.51	117.2	35.0
		MHW category IV	22.2	21.9	8.152	8.51	119.1	35.0
	25/06/2020	Control	18.2	18.5	8.176	8.55	112.5	35.6
		MHW category III	21.6	21.5	8.203	8.27	114.9	35.0
		MHW category IV	22.7	22.5	8.185	8.41	119.5	35.6
	26/06/2020	Control	18.2	18.4	8.084	8.26	107.9	35.0
		MHW category III	21.6	21.3	8.102	8.27	114.4	35.0
		MHW category IV	22.7	22.5	8.042	7.92	112.1	35.0
	27/06/2020	Control	18.2	18.4	8.084	8.35	109.9	36.0
		MHW category III	21.6	21.3	8.107	8.32	115.9	36.0
		MHW category IV	22.7	22.6	8.061	8.04	114.3	35.6
	28/06/2020	Control	18.2	18.4	8.095	8.50	112.1	36.4
		MHW category III	21.6	21.4	8.113	8.38	116.9	36.2
		MHW category IV	22.7	22.5	8.076	8.56	122.0	36.0
	29/06/2020	Control	18.3	18.5	8.091	8.49	112.7	37.0
		MHW category III	21.6	21.4	8.114	8.56	120.1	37.0
		MHW category IV	22.7	22.5	8.088	8.35	119.6	37.0
	30/06/2020	Control	18.3	18.4	8.125	8.66	114.5	36.8
		MHW category III	21.6	21.4	8.138	8.39	117.4	36.8
		MHW category IV	22.7	22.5	8.111	8.65	123.4	36.2
	01/07/2020	Control	18.3	18.4	8.097	8.20	107.6	35.6
		MHW category III	21.6	21.5	8.148	8.36	116.4	35.5
		MHW category IV	22.7	22.5	8.095	8.42	119.6	35.6
	02/07/2020	Control	18.3	18.4	8.126	8.67	114.2	36.0
		MHW category III	21.6	21.4	8.173	8.63	120.3	36.0
		MHW category IV	22.7	22.5	8.133	8.61	122.6	36.0
	03/07/2020	Control	18.3	18.3	8.122	8.30	109.5	37.0
		MHW category III	21.6	21.2	8.149	8.18	114.4	37.0
		MHW category IV	22.6	22.3	8.120	8.17	116.6	37.0
	04/07/2020	Control	18.3	18.5	8.110	8.64	113.2	35.0
		MHW category III	21.5	21.3	8.152	8.84	122.3	35.0
		MHW category IV	22.6	22.3	8.084	8.42	118.7	35.0
	05/07/2020	Control	18.4	18.5	8.175	8.60	113.0	35.6
		MHW category III	21.5	21.3	8.153	8.69	120.4	35.3
		MHW category IV	22.6	22.4	8.123	8.57	121.2	35.2
	06/07/2020	Control	18.4	18.6	8.123	8.33	110.0	36.0
		MHW category III	21.5	21.4	8.140	8.63	120.4	36.0
		MHW category IV	22.5	22.5	8.073	7.95	113.0	35.6
	07/07/2020	Control	18.4	18.6	8.098	8.35	110.3	36.0
		MHW category III	21.5	21.4	8.124	8.63	120.3	36.0
		MHW category IV	22.5	22.3	8.109	8.37	118.8	36.0
	08/07/2020	Control	18.4	18.6	8.026	7.91	104.6	36.0
		MHW category III	21.5	21.5	8.077	8.20	114.1	35.5

Recovery period		MHW category IV	22.5	22.4	8.041	8.07	114.7	36.0
	09/07/2020	Control	18.4	18.6	8.111	8.63	114.2	36.0
		MHW category III	21.1	21.2	8.206	8.69	120.7	36.0
		MHW category IV	21.9	22.1	8.095	7.95	112.2	36.0
	10/07/2020	Control	18.4	18.6	8.046	8.27	108.7	35.0
		MHW category III	20.6	20.7	8.124	8.92	122.1	35.0
		MHW category IV	21.4	21.4	8.100	8.22	114.1	35.0
	11/07/2020	Control	18.4	18.6	8.030	8.46	111.1	35.0
		MHW category III	20.2	20.4	8.084	8.43	114.7	35.0
		MHW category IV	20.8	20.9	8.070	8.10	111.4	35.0
	12/07/2020	Control	18.5	18.6	8.112	8.36	110.2	35.6
		MHW category III	19.8	20.0	8.156	8.94	120.7	35.2
		MHW category IV	20.2	20.3	8.111	8.27	112.2	35.0
	13/07/2020	Control	18.5	18.7	8.089	8.57	112.8	35.0
		MHW category III	19.4	19.6	8.158	8.83	118.2	35.0
		MHW category IV	19.6	19.8	8.127	8.54	114.8	35.0
	14/07/2020	Control	18.5	18.6	8.098	8.39	111.0	36.0
		MHW category III	18.9	19.2	8.140	8.45	112.9	36.0
		MHW category IV	19.1	19.4	8.108	8.08	108.5	36.0
	15/07/2020	Control	18.5	18.7	8.018	8.41	110.7	35.0
		MHW category III	18.5	18.9	8.048	8.65	114.3	35.0
		MHW category IV	18.5	19.1	7.980	7.77	103.0	35.0

Supplementary Table 2 Multiple comparison analysis of Pheophytin *a* concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Contrast	estimate	SE	df	t-ratio	p-value
Control - MHW III	-69.64	28.9	12	-2.413	0.0778
Control - MHW IV	-5.51	32	12	-0.172	0.9837
MHW III - MHW IV	64.13	30.8	12	2.084	0.1347

Supplementary Table 3 Multiple comparison analysis of Antheraxanthin concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Contrast	estimate	SE	df	t-ratio	p-value
Control - MHW III	-7.47	3.06	12	-2.445	0.0735
Control - MHW IV	-3.42	3.39	12	-1.011	0.5838
MHW III - MHW IV	4.05	3.26	12	1.243	0.4514

Supplementary Table 4 Multiple comparison analysis of Total Carotenoids concentrations in *Zostera noltei* leaves. Statistical significance at p-value<0.05.

Contrast	estimate	SE	df	t-ratio	p-value
Control - MHW III	-45.43	17.6	12	-2.578	0.0584
Control - MHW IV	-6.92	19.5	12	-0.355	0.9334
MHW III - MHW IV	38.51	18.8	12	2.05	0.1423

Supplementary Table 5 Macrofauna species composition in all experimental treatments. N: total number of individuals in the community; S: total number of species; n_i : the abundance of the i -th specie; p_i : relative abundance.

Class	Taxa	Control	MHW category III	MHW category IV	n_i	p_i
Bivalvia	<i>Abra alba</i>	0	3	0	3	0.012
Bivalvia	<i>Cerastoderma glaucum</i>	6	7	4	17	0.069
Bivalvia	<i>Nucula</i> sp.	0	1	0	1	0.004
Bivalvia	<i>Parvicardium exiguum</i>	1	0	0	1	0.004
Bivalvia	<i>Ruditapes philippinarum</i>	3	2	0	5	0.020
Gastropoda	<i>Steromphala cineraria</i>	0	0	2	2	0.008
Gastropoda	<i>Nucella lapillus</i>	3	9	6	18	0.073
Gastropoda	<i>Peringia ulvae</i>	51	73	42	166	0.672
Malacostraca	<i>Carcinus maenas</i>	12	13	9	34	0.138
	N	76	108	63	247	1
	S	6	7	5		